AEROSPACE MATERIAL SPECIFICATION

AMS2750™	REV. H

Issued 1980-04
Reaffirmed 1994-04
Revised 2024-07

Superseding AMS2750G

Pyrometry

RATIONALE

AMS2750H results from a Two-Year Review and update of this specification with changes to Definitions (see 2.4.21, 2.4.25, 2.4.36, 2.4.47, and 2.4.84); General Sensor Requirements (see Table 3); Sensor Calibration (see 3.1.4.6 and Table 5); SAT and TUS Sensor Reuse (see 3.1.7.4 and 3.1.7.5); Base Metal Load Sensors (see 3.1.10.3); General Instrumentation Requirements (see 3.2.1.4.1, 3.2.1.5, and Table 7); Control, Recording, and Over-Temperature Instruments (see 3.2.3.4, 3.2.3.5, 3.2.3.16, and 3.2.3.18); Instrumentation Calibration Results and Records (see 3.2.5.1); General Instrument Correction and Modification Offset Requirements (see 3.2.6.1.2 and 3.2.6.1.8); Thermal Processing Equipment (see 3.3.7); General SAT Requirements (see Table 11, Table 12, and 3.4.1.2.1); Performing an SAT (see 3.4.2.2); Alternate SAT Frequency (see 3.4.8.3); SAT Waiver (see 3.4.9.6.3); SAT Difference Pass/Fail Requirements (see 3.4.10.4); Comparison SAT (see 3.4.11.1.e); Parts furnace class, instrument type, and TUS internal (see Table 15); Raw material furnace class, instrument type, and TUS internal (see Table 16); Initial TUS Temperatures (see 3.5.2.4); TUS Requirements for Batch Furnaces, Salt Baths, Controlled Temperature Liquid Baths, and Fluidized Bed Furnaces (Excluding Controlled Temperature Quench Baths) (see Table 17); TUS Data Collection (see 3.5.10.1 and 3.5.10.2); Relocation of Hot or Cold Recording Sensors for Type A and C Instrumentation (see 3.5.15.2); Radiation Survey (see 3.5.17, 3.5.17.1, and 3.5.17.2); TUS Interval Deviations (see 3.5.18); Rounding (see 3.8); and Quality Assurance Provisions (see 4.2, 4.4, and Table 22). Deleted Zener Voltage Reference (was 3.5.87).

1. SCOPE

1.1 This specification covers pyrometric requirements for equipment used for the thermal processing of metallic materials. Specifically, it covers temperature sensors, instrumentation, thermal processing equipment, correction factors and instrument offsets, system accuracy tests, and temperature uniformity surveys. These are necessary to ensure that parts or raw materials are heat treated in accordance with the applicable specification(s).

1.2 This specification may be used in other non-heat-treating applications when specified.

1.3 This specification is not applicable to heating or to intermediate thermal processing unless otherwise specified.

1.4 This specification applies to laboratory furnaces to the extent specified in 3.6.

2. APPLICABLE DOCUMENTS

The issue of the following documents in effect on the date of the purchase order forms a part of this specification to the extent specified herein. The supplier may work to a subsequent revision of a document unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply.

SAE Executive Standards Committee Rules provide that: “This report is published by SAE to advance the state of technical and engineering sciences. The use of this report is entirely voluntary, and its applicability and suitability for any particular use, including any patent infringement arising therefrom, is the sole responsibility of the user.”

SAE reviews each technical report at least every five years at which time it may be revised, reaffirmed, stabilized, or cancelled. SAE invites your written comments and suggestions.

Copyright © 2024 SAE International

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, or used for text and data mining, AI training, or similar technologies, without the prior written permission of SAE.

TO PLACE A DOCUMENT ORDER: **Tel: 877-606-7323 (inside USA and Canada)**
Tel: +1 724-776-4970 (outside USA)
Fax: 724-776-0790
Email: CustomerService@sae.org
http://www.sae.org

SAE WEB ADDRESS:

For more information on this standard, visit
https://www.sae.org/standards/content/AMS2750H/

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris

Licensee=Marshall of Cambridge Aero - Cambridge/9992345108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

AS7766 Terms Used in Aerospace Metals Specifications

2.2 ASTM Publications

Available from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959, Tel: 610-832-9585, www.astm.org.

ASTM E29 Using Significant Digits in Test Data to Determine Conformance with Specifications

ASTM E207 Standard Test Method for Thermal EMF Test of Single Thermoelement Materials by Comparison with a Reference Thermoelement of Similar EMF-Temperature Properties

ASTM E220 Calibration of Thermocouples by Comparison Techniques

ASTM E230 Temperature-Electromotive Force (EMF) Tables for Standardized Thermocouples

ASTM E608 Mineral-Insulated, Metal-Sheathed Base Metal Thermocouples

ASTM E1137 Industrial Platinum Resistance Thermometers

ASTM E1751 Standard Guide for Temperature Electromotive Force (emf) Tables for Non-Letter Designated Thermocouple Combinations

ASTM MNL7 Presentation of Data and Control Chart Analysis

ASTM MNL12 Use of Thermocouples in Temperature Measurement

2.3 IEC Publications

Available from IEC Central Office, 3, rue de Varembe, P.O. Box 131, CH-1211 Geneva 20, Switzerland, Tel: +41 22 919 02 11, www.iec.ch.

IEC 60751 Industrial Platinum Resistance Thermometers and Platinum Temperature Sensors

ISO/IEC 17025 General Requirements for the Competence of Testing and Calibration Laboratories

2.4 Definitions

Terms used in AMS2750 are defined in AS7766 and as follows:

2.4.1 ACCURACY

The maximum deviation of the instrument or sensor being tested from the values of a traceable standard.

2.4.2 ADJUSTMENT

Any change to an instrument’s parameters.

2.4.3 AUTOCLAVE

An oven capable of operating at pressures higher than atmospheric pressure (nominally 760 mm Hg), commonly used in the processing of materials. It may be pressurized with steam, compressed air, or inert gas.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.4 BASE METAL SENSOR

Sensor whose thermoelements are composed primarily of base metals and their alloys. Examples of base metal sensors include Types E, J, K, N, M, and T.

2.4.5 BATCH FURNACE

A furnace where parts or raw material are stationary during the soak.

NOTE: Some batch furnaces may oscillate material within a stationary work zone.

2.4.6 BIAS or INPUT SHIFT

The act of making an adjustment to an instrument to add, remove, or alter an offset.

2.4.7 BIMONTHLY

--`,`,,,,``,,`,`````,``,,``,,`-``

See FREQUENCY.

2.4.8 BIWEEKLY

See FREQUENCY.

2.4.9 CALIBRATION

An assessment of the accuracy of a sensor or an instrument to a traceable standard sensor and/or field test or standard instrument, based on one or more measurements, and potentially adjusting an instrument and/or compiling a deviation chart for a sensor or instrument in order to ensure compliance with requirements.

2.4.10 CONTINUOUS FURNACE

A furnace where parts or raw material are conveyed continuously or semi-continuously from the charge area to the discharge area. Examples include: bump furnace, shaker furnace, belt furnace, roller furnace, and rotary hearth furnace.

2.4.11 CONTROL INSTRUMENT

An instrument connected to a control sensor used to control the temperature of thermal processing equipment. The instrument may or may not also record temperature data.

2.4.12 CONTROL SENSOR

A sensor connected to a control instrument on thermal processing equipment, the temperature of which may or may not be recorded.

2.4.13 CONTROL ZONE

A portion of the working zone in thermal processing equipment having a separate sensor, instrument, and heating or cooling system to control its temperature. This portion of the thermal processing equipment is independently controlled.

2.4.14 CONTROLLED TEMPERATURE LIQUID BATH

A furnace containing a liquid that is heated to the desired heat-treat temperature. Parts and raw material are normally immersed in the liquid.

2.4.15 CONTROLLER

A digital or mechanical device that controls the temperature of thermal processing equipment (e.g., furnace control instruments, quench mechanical thermostat, freezer pressure controls, etc.).

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.16 CORRECTION FACTOR

The number of degrees, determined from the most recent calibration, that must be added to, or subtracted from, the temperature reading of a sensor, or an instrument, or a combination thereof (system) to obtain true temperature. The correction factors of sensors and instruments are usually kept separately and added together algebraically when a combination is used. Correction factor is the algebraic opposite of deviation (error).

2.4.17 DATA ACQUISITION SYSTEM

An instrument system used to automatically collect and store process data as an electronic record; for example, a Programmable Logic Controller (PLC).

2.4.18 DEVIATION/ERROR

In the context of this specification, the difference between the uncorrected indicated temperature and the true temperature (Indicated Temperature - True Temperature = Deviation/Error).

2.4.19 DIGITAL INSTRUMENT

An instrument that records process measurements in a digital (numeric) display format or an instrument that prints both the scale (graph) and trend line simultaneously. Examples include recorders with pre-printed scales and printed tabular data meeting 3.2.1.4 data collection intervals and recorders that create scales and trend line or tabular data and also display indicated temperature.

2.4.20 ELECTRONIC RECORD

Any combination of text, graphics, data, audio, pictorial, or other information representation in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system.

2.4.21 EXPENDABLE SENSORS

Sensors where any portion of the thermoelements are exposed to the thermal process equipment environment. Sensors with braided fiberglass insulation or plastic insulation are not designed to be inserted into a closed protection tube as the confined space causes drift when the additives (e.g., Teflon, color dies, and other chemicals) burn off. Therefore, these sensors are expendable even when installed into a closed protection tube.

2.4.22 EXTENSION WIRE

Wire used for transmitting an unmodified signal from the sensor to the instrumentation system. Wire is generally of the same sensor type, except for some sensor types that allow compensated extension wire.

2.4.23 FIELD TEST INSTRUMENT

An instrument meeting the requirements of Table 7 that has calibration traceable to a secondary standard instrument or better and is used to conduct on-site calibrations or tests of thermal processing equipment.

2.4.24 FLUIDIZED BED FURNACE

A furnace that contains a medium that becomes suspended or fluidized due to atmosphere gas or products of combustion passing upward through the medium. Parts and raw material are normally immersed in the fluidized medium.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.25 FREQUENCY (INTERVAL)

The calendar days from the day/date a calibration, test, or sensor replacement was performed and the next day/date a calibration, test, or sensor replacement is due (inclusive). In the context of this specification, the following shall apply:

Frequency	Equal to Once Every	Shall Be Completed On or Before
Weekly	Week	The same day of the week every week
Biweekly	2 weeks	The same day of the week every 2 weeks
Monthly	Month	The same date of the month every month(1)
Bimonthly	2 months	The same date of the month every 2 months(1)
Quarterly	3 months	The same date of the month every third month(1)
Semiannually	6 months	The same date of the month every 6 months(1)
Annually	Year	The same date of the year every year(1)

(1) If the next calibration, test, or sensor replacement is due on a calendar date not contained in that month, then the last day of that calendar month shall be used for the next calibration (e.g., a monthly calibration, test, or sensor replacement performed on January 31 is due on the last day of February).

2.4.26 FURNACE

Equipment used for the thermal processing of parts or raw material. The terms “furnace” and “oven” can be used interchangeably.

2.4.27 HEAT SINK

A mass of material with an embedded sensor or sensors that supplies temperature data of that mass to recording instruments.

2.4.28 INTERVAL

See FREQUENCY.

2.4.29 LABORATORY THERMAL PROCESSING EQUIPMENT

Equipment used exclusively for thermal processing of samples, specimens, or test parts as required by materials and processing specifications.

2.4.30 LOAD SENSOR

Sensor that is attached to, or in contact with, parts or raw material, a representation of parts or raw material, or is buried in the load of parts (e.g., fasteners) or raw material and which supplies temperature data of the parts or raw material to recording instruments that may be used to control the sequence of the production process, such as the start of soak.

2.4.31 MATERIAL PRODUCER (METALLIC)

The manufacturer of raw material as stated in 2.4.53.

2.4.32 MAXIMUM PERMITTED ERROR

A tolerance band for the thermal electric response expressed in degrees or percentages. Maximum permitted error provides a tolerance within which various types of sensors shall conform to standard sensor reference tables, or equivalent.

2.4.33 MEASURING JUNCTION

The location of a sensor where the wire elements are joined together to complete a measurement circuit, which is used to measure an unknown temperature. Also called the “hot junction.”

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.34 MULTIPLE ZONED FURNACES

Furnaces with multiple separate temperature control zones.

2.4.35 NOBLE METAL SENSOR

Sensor whose thermoelements are composed primarily of noble metals (e.g., platinum/platinum-rhodium) and their alloys. Examples of noble metal sensors include Types R, S, B, and RTDs.

2.4.36 NONEXPENDABLE SENSORS

Sensors having no portion of the thermoelements exposed to the thermal process equipment environment. Sensors with ceramic insulators over bare wire inserted into a closed protection tube to prevent exposure to the thermal process equipment environment are considered nonexpendable sensors.

2.4.37 NONMETALLIC MATERIALS

In the context of this specification, this term refers to the curing of composite or adhesive bonded assemblies that are typically processed in autoclaves, air ovens, or heated presses.

2.4.38 OFFSET

Any manual or electronic adjustment to an instrument made to alter either the desired set point or the displayed value of the instrument's calculated temperature. Manufacturer specific terminology may also include "bias," "input shift," etc.

2.4.38.1 CORRECTION OFFSET

Manual or electronic adjustment of an instrument to compensate for known errors of the measurement system (instrument, extension wire/connectors, sensor) to make the system more accurate.

2.4.38.2 MODIFICATION OFFSET

Manual or electronic adjustment of an instrument to compensate for known conditions such as, but not limited to, a skewed TUS result or control thermocouple placement in a retort or muffle.

2.4.39 OVEN

Equipment used for the thermal processing of materials and parts. The terms "oven" and "furnace" can be used interchangeably.

2.4.40 OVER-TEMPERATURE INSTRUMENTATION

An independent sensor and instrument combination installed in the thermal processing equipment that is used to detect any over-temperature occurrence and generate an alarm and/or cut back or shut down heat input. The purpose for this control is to protect parts or raw material and/or the thermal processing equipment from overheating. Integrated control/recording/over-temperature instruments are permitted provided it can be demonstrated that the over-temperature instrument/module of an integrated system is separated from the furnace control/recording system.

2.4.41 PARTS

Usually identified by a part number, produced from raw material in accordance with the requirements of an engineering drawing, and usually tested by nondestructive techniques only. Parts are heat treated, by or for a fabricator, in accordance with a drawing, purchase order, fabrication order, or heat-treat specification.

NOTE: The cognizant engineering organization has the authority to assign the terms "parts" or "raw material."

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.42 PREVENTIVE MAINTENANCE PROGRAM or PM PROGRAM

A program for evaluating, taking corrective action as required, and documenting the condition of items that have potential to adversely affect thermal processing equipment conformance to any requirement of this specification. Frequency of PM checks is established based on experience to ensure that no major problems occur between periodic PMs.

2.4.43 PRIMARY STANDARD INSTRUMENT

An instrument that is calibrated directly against a reference standard instrument and meeting the requirements of Table 7.

2.4.44 PRIMARY STANDARD SENSOR

Sensor calibrated directly against a reference standard and meeting the requirements of Table 1.

2.4.45 PROCESS CHART RECORDER

See RECORDING INSTRUMENT.

2.4.46 PROGRAMMABLE LOGIC CONTROLLER (PLC)

A digital computer control system that continuously monitors the state of input devices and makes decisions based upon a programmed input (recipe) to control the state of output devices.

2.4.47 QUALIFIED OPERATING TEMPERATURE RANGE

The nominal set point temperature range of thermal processing equipment where temperature uniformity has been tested within a qualified work zone and found to be compliant with required tolerances (i.e., qualified operating temperature range = nominal range of TUS set points).

2.4.48 QUALIFIED WORK ZONE

The portion of a thermal processing equipment volume where temperature variation conforms to the required uniformity tolerance within the qualified operating temperature range as defined by the placement of sensors during the most recent temperature uniformity survey.

2.4.49 QUALITY ORGANIZATION APPROVAL

--`,`,,,,``,,`,`````,``,,``,,`-``,,`,,,,-``,,`,,,,---

Objective evidence of review and acceptance or rejection of a calibration or test as defined by a documented process within the user’s quality system, which also defines any delegation of this approval.

2.4.50 QUENCH SYSTEM

A system that provides rapid cooling, usually accomplished using oil, water, water/polymer mixtures, or gaseous mediums.

2.4.51 RADIATION SURVEY

Initial survey of aluminum alloy thermal processing equipment used above 800 °F or 427 °C when the heat source (e.g., electrical elements or gas tubes) is exposed to the qualified work zone or only separated by a metal baffle.

2.4.52 RADIATION SURVEY SENSOR

A TUS sensor, typically a base metal (Types E, J, K, N, M, and T) sensor, used in conjunction with a test panel for determining the heating characteristics of furnaces used for solution heat treating aluminum alloys.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.53 RAW MATERIAL

Usually includes, but is not limited to, such items as sheet, plate, wire, rod, bar, forgings, castings, and extrusions. Raw material is usually identified by a heat or lot number and is usually tested destructively for acceptance. Raw material is heat treated, by or for a material producer, in accordance with a process or material specification.

NOTE: The cognizant engineering organization has the authority to assign the terms “parts” or “raw material.”

2.4.54 RAW MATERIAL FURNACES

Equipment used in accordance with a process or material specification to process raw material.

2.4.55 RECORDING INSTRUMENT

An instrument connected to a controlling, load, and/or recording sensor that documents process equipment temperature data and generates a permanent process record. Examples are a chart recorder, electronic data recorder, or a data acquisition system.

2.4.56 RECORDING SENSOR

A sensor that is connected to a recording instrument or connected to a control instrument of an integrated control/recording system.

2.4.57 RECURRENT TEMPERATURE PATTERN

Cycling of furnace temperature due to operation of the temperature control instrument.

2.4.58 REFERENCE STANDARD INSTRUMENT

A standard test instrument that has been calibrated by NIST or other internationally recognized standards organization meeting Table 7 requirements.

2.4.59 REFERENCE STANDARD SENSOR (NOBLE METAL)

A noble metal standard sensor that has been calibrated by NIST or other internationally recognized standards organization meeting Table 1 requirements.

2.4.60 REFRACTORY SENSOR

A sensor whose thermoelements are composed primarily of refractory metals (e.g., Tungsten, Rhenium, Tantalum, Niobium, and Molybdenum) and their alloys.

Example: Type C sensors.

2.4.61 REFRIGERATION EQUIPMENT

A compartment, cabinet, or room that may be held below room temperature and >32 °F or 0 °C (refrigerator) or ≤32 °F or 0 °C (freezer), depending on the temperature range of use. This equipment may be used for retarding or advancing metallurgical transformation or for storage of metallic materials.

2.4.62 RESIDENT SAT SENSOR

A test sensor that remains resident in the test location between system accuracy tests.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.63 RESISTANCE TEMPERATURE DEVICE (RTD)

A device (for example, PT100, PRT, etc.) that produces a change in resistance across an element in response to the temperature at the element (usually in the tip).

2.4.64 RETORT FURNACE

A furnace that contains a retort or muffle that isolates the parts or raw material being heat treated from the heating elements. The furnace normally surrounds the retort.

2.4.65 SALT BATH

A furnace containing molten salt that is heated to the desired heat-treat temperature. Parts or raw material are normally immersed in the molten salt.

2.4.66 SECONDARY STANDARD TEST INSTRUMENT

An instrument calibrated directly against a primary standard or reference standard meeting Table 7 requirements and which is operated in a controlled test environment.

2.4.67 SECONDARY STANDARD TEST SENSOR

A sensor calibrated directly against a primary standard test sensor meeting the requirements of Table 1.

2.4.68 SEMI-CONTINUOUS FURNACE

See CONTINUOUS FURNACE.

2.4.69 SENSOR OR TEMPERATURE SENSOR

In the context of this specification, a device designed to detect or measure temperature (e.g., thermocouple, RTD, etc.).

2.4.70 SPECIAL LIMITS OF ERROR SENSOR WIRE

Sensors and extension wires whose initial calibration accuracy meets or exceeds the requirements of ASTM E230, Tables 2 and 3 for special tolerances.

2.4.71 STABILIZATION (also referred to as EQUALIZATION, EQUILIBRIUM, STEADY STATE, or SOAKED CONDITION)

Equipment stabilization occurs when all control and recording sensors are within the allowable temperature uniformity survey tolerance span and controllers are cycling and/or maintaining the desired temperature in each zone. Temperature uniformity survey stabilization occurs when all temperature uniformity survey sensors have reached the desired uniformity range and are not exhibiting a continual upward or downward trend away from the set point during and after the stabilized period of the temperature uniformity survey.

2.4.72 SYSTEM ACCURACY TEST (SAT)

An assessment of the sum of the combined errors or correction factors of the sensor, extension wire (and connectors), and instrument to ensure compliance with Table 11 or Table 12 requirements.

2.4.72.1 COMPARISON SAT

An assessment by comparison of the difference between the readings of the thermal process equipment sensor system being tested (sensor, extension wire, and instrument) and the corrected reading of the test sensor system (test sensor, extension wire, and field test instrument) after test sensor and field test instrument correction factors are applied (see 3.4.7). Also referred to as a “probe check.”

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.72.2 ALTERNATE SAT

A mathematical assessment of the sum of the errors or correction factors of the thermal processing equipment sensor and the calibration error or correction factor of the connector, extension wire, and instrument channel (see 3.4.8).

2.4.72.3 SAT WAIVER

Additional requirements and comparisons to be made when the comparison or alternate SAT methods are not performed (see 3.4.9).

2.4.73 SYSTEM ACCURACY TEST SENSOR

A calibrated and traceable sensor meeting the requirements of Table 1 used for an SAT.

2.4.74 TEMPERATURE OVERSHOOT

When any temperature sensor exceeds the upper temperature tolerance as defined by the applicable thermal processing equipment class as stated in Table 8.

2.4.75 TEMPERATURE SENSOR PASS-THROUGH

Installed wiring connecting sensors (typically thermocouples) inside the thermal processing equipment to the instrumentation outside, usually with sockets, jacks, or terminals at each end.

2.4.76 TEMPERATURE UNIFORMITY

The temperature variation (usually expressed as ± degrees) within the qualified work zone with respect to set point temperature. For retort furnaces where a sensor in the retort is used to control temperature, the temperature variation is with respect to the sensor in the retort and not to the furnace set point temperature. The requirement is established by the required thermal processing equipment class as stated in Table 8.

2.4.77 TEMPERATURE UNIFORMITY RECORDER

Independent digital recording instrument meeting the requirements of Table 7 for a field test instrument and used to perform temperature uniformity surveys.

2.4.78 TEMPERATURE UNIFORMITY SENSOR

A calibrated and traceable sensor meeting the requirements of Table 1.

2.4.79 TEMPERATURE UNIFORMITY SURVEY (TUS)

An assessment of the temperature variation within the qualified work zone of thermal processing equipment prior to and after stabilization using a field test instrument (TUS recorder) meeting Table 7 requirements and sensors meeting the requirements of Tables 1, 17, and 18, as applicable.

2.4.80 THERMAL PROCESSING

Any process in which parts or raw material are exposed to controlled heating, soaking, or cooling to achieve the specified properties or condition of the parts or raw material for which there is no exception stated in 1.3.

2.4.81 THERMAL PROCESSING EQUIPMENT

A term used to refer to any vessel (such as autoclave, furnace, oven, quench and refrigeration equipment, liquid bath, heated press, etc.) used to process parts or raw material at controlled temperatures.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

2.4.82 THERMOCOUPLE

A temperature sensor consisting of two wires (thermoelements) of dissimilar thermoelectric characteristics connected at a measuring junction. An EMF is developed between the two junctions in proportion to the temperature gradient.

2.4.83 TRACEABLE OR TRACEABILITY

The ability to relate measurement results through an unbroken chain traceable to the International System of Units (SI) through internationally recognized standards organizations such as, but not limited to:

National Institute of Standards and Technology (NIST)
National Physical Laboratory (NPL)
Physikalisch-Technische Bundesanstalt (PTB)
Swedish National Authority for Testing, Inspection, and Metrology
China National Calibration Technology Specification (CNAS)
National Institute of Advanced Industrial Science and Technology (AIST)
Instituto Nacional de Metrologia, Normalização e Qualidade Industrial (INMETRO)
Bureau International des Poids et Mesures (BIPM)

2.4.84 USE (OF A SENSOR)

One cycle of heating or cooling upon the sensor being placed into service (see Table 5 note 5, 3.1.4.2, 3.1.7.2, 3.1.7.5, and 3.1.11.1).

2.4.85 VACUUM FURNACE

A furnace that processes parts or raw material at any pressure lower than atmospheric pressure (nominally 760 mm Hg) during soak.

2.4.86 WIRELESS TRANSMITTER

A device for sending electromagnetic waves; the part of a broadcasting apparatus that generates and modulates the radio frequency current and conveys it to a receiver.

--`,,,,,``,,`,`````,``,,``,,`-``,,`,,,,-``,,`,,,,---

3. TECHNICAL REQUIREMENTS

3.1 Temperature Sensors

3.1.1 General Sensor Requirements

3.1.1.1 Unless specifically noted, the requirements defined in this specification shall apply to all sensors.

3.1.1.2 All sensors shall comply with the requirements of Table 1. Other sensors that possess equal or better calibration accuracy may be used. Thermocouple composition shall comply with the requirements of ASTM E230 or ASTM E1751 and Table 2.

3.1.1.3 Resistance temperature devices (RTDs) shall be noble metal, shall comply with the requirements of Table 1, and shall be considered nonexpendable.

3.1.1.4 Sensors may be made either from bare or coated wire or mineral insulated/metal sheathed (MIMS) cable as described in Tables 2 and 3.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.1.1.5 Measuring junctions shall be made by either of the following methods:

- Any combination of twisting and/or welding the thermoelements provided there is no addition of filler metal (including ungrounded and grounded MIMS).

- Spot welding the thermoelements directly to a part, simulated part, or heat sink is permitted for temperatures ≤2000 °F or 1100 °C.

Table 1 - Sensors and sensor calibration(12)

<table>
<tr><th rowspan="2">Sensor</th><th rowspan="2">Use</th><th rowspan="2">Type(1)(10)</th><th colspan="2">Calibration</th><th rowspan="2">Calibration Accuracy(2)</th></tr>
<tr><th>Interval</th><th>Standard</th></tr>
<tr><td>Reference standard</td><td>Primary standard calibration(5)</td><td>B, R, S</td><td>Before first use; 5 years</td><td>NIST(4) reference standard</td><td>Refer to NIST or equivalent calibration report</td></tr>
<tr><td>Primary standard</td><td>Secondary standard calibration(6)</td><td>B, R, S</td><td>Before first use; 3 years</td><td>Reference standard(5)</td><td>Type R, S: ±1.0 °F or ±0.6 °C or ±0.1%
Type B: ±0.25%</td></tr>
<tr><td>Secondary standard(7)</td><td>Sensor calibration</td><td>B, R, S, RTD(10)</td><td>Before first use; 2 years</td><td>Primary standard(6)</td><td rowspan="5">Type R, S: ±1.0 °F or ±0.6 °C or ±0.1%(13)
Type B: ±0.25%(13)
Refractory(9)
Base metal: ±2.0 °F or ±1.1 °C or ±0.4%(11)
RTD(10)</td></tr>
<tr><td colspan="2">System accuracy test(3)</td><td rowspan="2">B, R, S, RTD, base metal, and refractory</td><td rowspan="2">Before first use(3)</td><td rowspan="2">Primary or secondary standard(8)</td></tr>
<tr><td colspan="2">Temperature uniformity survey(3)</td></tr>
<tr><td>Control and recording(3)</td><td>Installation in thermal processing equipment</td><td rowspan="2">B, R, S, RTD, base metal, and refractory</td><td>Before first use(3)</td><td rowspan="2">Primary or secondary standard(8)</td></tr>
<tr><td>Load(3)</td><td>Sensing part or raw material temperature</td><td>Before first use(3)</td></tr>
</table>

(1) Sensors of equal or better calibration accuracy are acceptable.
(2) Percent of reading or correction factor in °F or °C, whichever is greater.
(3) Sensor recalibration and reuse requirements are provided in Table 5.
(4) NIST or other internationally recognized standards organization.
(5) A reference standard sensor together with a primary standard instrument shall be used to calibrate primary standard sensors.
(6) A primary standard sensor together with a primary standard instrument shall be used to calibrate secondary standard sensors.
(7) Use shall be limited to the calibration of control, recording, and load sensors, and SAT and TUS sensors.
(8) A primary or secondary standard sensor together with a primary or secondary standard instrument shall be used to calibrate these sensors.
(9) For refractory sensors Type C: ±8.0 °F or ±4.4 °C or ±1%, the sensor correction factor shall be used for all applications.
(10) RTDs, when used, shall be platinum type and meet Class/Grade A tolerances given in ASTM E1137 or IEC 60751. This does not apply to RTDs used in conjunction with a refrigeration equipment controller.
(11) For temperatures <32 °F or <0 °C for Types K, E, and T only, calibration accuracy shall meet the following:
- Type K: -328 to 32 °F, ±4.0 °F or -200 to 0 °C, ±2.2 °C, or ±2.0% for either, whichever is greater.
- Type E (MIMS): -328 to 32 °F, ±4.0 °F or -200 to 0 °C, ±2.2 °C, or ±2.0% for either, whichever is greater.
- Type E (all other): -328 to 32 °F, ±3.0 °F or -200 to 0 °C, ±1.7 °C, or ±1.0% for either, whichever is greater.
- Type T: -328 to 32 °F, ±1.8 °F or -200 to 0 °C, ±1.0 °C, or ±1.5 % for either, whichever is greater.

(12) Sensors and sensor materials are normally supplied to meet the tolerances specified in the table for temperatures above 32 °F or 0 °C. The same materials, however, may not fall within the tolerances for temperatures below 32 °F or 0 °C. Two separate sensors may be required, one for each range.
(13) When correction factors are used during production:
- Type B sensors shall meet a calibration accuracy of ±0.5%,
- Types R and S sensors shall meet calibration accuracy of ±2.7 °F or ±1.5 °C or ±0.25%, whichever is greater.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 2 - Sensor and extension wire/connector

Sensor					Extension Wire		
Classification	Type	Positive Element Composition Nominal wt. %	Negative Element Composition Nominal wt. %	Element Color Code(1)	Wire Code Positive/Negative	Jacket Color Code(1)	Connector Color Code(1)
Base Metal	J	Fe	55Cu/45Ni	White/Red	JPX/JNX	Black	Black
	E	90Ni/10Cr	55Cu/45Ni	Purple/Red	EPX/ENX	Purple	Purple
	K	90Ni/10Cr	95Ni/5, Al + Si	Yellow/Red	KPX/KNX	Yellow	Yellow
	N	84.5Ni/14Cr/1.5Si	95.4Ni/4.5Si/0.1Mg	Orange/Red	NPX/NNX	Orange	Orange
	M(3)	82Ni/18Mo	Ni	Yellow/Red	KPX/KNX	Yellow	Yellow
	T	Cu	55Cu/45Ni	Blue/Red	TPX/TNX	Blue	Blue
Noble Metal(2)	R	87Pt/13Rh	PT	Black/Red	RPX/RNX or SPX/SNX	Green	Green
	B	70Pt/30Rh	94Pt/6Rh	Gray/Red	BPX/BNX	Gray	Gray
	S	90Pt/10Rh	PT	Black/Red	RPX/RNX or SPX/SNX	Green	Green
Refractory(2)	C	95W/5Re	74W/26Re	Green/Red	CPX/CNX	Red	Red

(1) All color codes stated are in accordance with ASTM E230. Color codes in accordance with other internationally recognized standards organizations are acceptable.

(2) Most base metal extension wires have the same nominal composition as the sensor wires with which they are intended to be used, whereas the compensating extension wires for noble metal (Types S, R, and B) or refractory metal sensors (Type C) are usually of a different, more economical composition whose relative thermoelectric properties as a pair nonetheless closely approximate those of the noble metal or refractory metal sensors with which they are to be used over a limited temperature range.

(3) The designation “Type M” may not be recognized by all international standards. Type K extension wire and connectors shall be used for Type M sensors.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris

Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 3 - Sensor classification

Sensor Cover	Sensor Classification(1)
Fiberglass, plastic, or trade name	Expendable
Multiple hard fired ceramic beads	Expendable
Metal over-braid	Expendable
Shielded from the process atmosphere by a closed metal/ceramic protective tube	Nonexpendable
Mineral insulated; metal sheathed (MIMS)	Nonexpendable
Mineral insulated; metal sheathed (MIMS) with an exposed measuring junction	Expendable

(1) See definitions for expendable and nonexpendable sensors.

3.1.2 Sensor Temperature Range of Use

Guidelines for sensor temperature range usage can be found in ASTM MNL12, ASTM E230, ASTM E608, ASTM E1137, ASTM E1751, IEC 60751, or other internationally recognized standards, and the sensor supplier.

3.1.3 Extension Wire and Connectors

3.1.3.1 Extension wire shall be of the same nominal composition as the sensor and instrumentation used, except when compatible compensating extension wire is allowed (e.g., noble metals). Extension wire shall meet the requirements of Table 2.

3.1.3.2 Extension wire shall not be spliced other than using a compatible connector.

3.1.3.3 Connectors, plugs, jacks, and terminal strips are permitted if they are the compatible type, i.e., they have thermoelectric properties conforming to the characteristics of the corresponding sensor type.

3.1.3.4 Wireless transmitters may be used as an alternative to extension wire.

3.1.4 Sensor Calibration

3.1.4.1 The sensor calibration technique shall comply with ASTM E207, ASTM E220, or other internationally recognized standards.

3.1.4.2 Sensors shall be calibrated before their first use.

3.1.4.3 Sensor calibration intervals, whether based on time, number of uses, or temperatures, are the maximums permitted.

3.1.4.4 Users shall have procedures that control the replacement of thermal process equipment sensors including limits on maximum life and/or number of uses based on supporting data such as, but not limited to, SAT, TUS, recalibration data, and/or trend analysis.

3.1.4.5 Sensors shall be calibrated or recalibrated at, or lower than, the lowest temperature of use and at, or higher than, the highest temperature of use. Sensors used at a single temperature may be calibrated at the single temperature of use.

3.1.4.6 Intervals between calibration or recalibration temperatures provided by the calibration agency shall not exceed 250 °F or 140 °C for all sensors. The fixed point calibration method in ASTM MNL12 or other internationally recognized standards may be used. No matter what standard is used, the calibration agency shall provide additional calibration points that do not exceed 250 °F or 140 °C for all sensors.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.1.4.7 Extrapolation of calibration correction factors above the highest calibration temperature and below the lowest calibration temperature is prohibited by any calibration source except NIST or other internationally recognized standards organization.

3.1.4.8 Interpolation of correction factors between two known calibration points is permitted using the linear method.

3.1.4.9 Alternatively, the correction factor of the nearest calibration point shall be used.

3.1.4.10 Whichever method is used shall be defined and applied consistently.

3.1.4.11 For recalibration of sensors when permitted in Table 5, it is acceptable to use either the date of recalibration or the date of first use following recalibration as the beginning of the calibration period. Procedures shall identify how the practice is applied and documented to ensure compliance.

3.1.5 Wire/Cable Rolls

3.1.5.1 Calibrated expendable or nonexpendable sensors made from rolls may be used in lieu of individually calibrated sensors.

3.1.5.2 The maximum length of wire/cable in a roll at the time of calibration shall comply with Table 4.

Table 4 - Maximum permitted length of wire/cable in a roll

Sensor	Maximum Permitted Length of Wire/Cable in a Roll
Primary standard	200 feet or 60 m
All other noble metal	2000 feet or 610 m
All other base metal	5000 feet or 1525 m
Refractory	2000 feet or 610 m

3.1.5.3 Rolls shall be sampled and calibrated at both ends. The individual correction factors from each end shall be within the requirements of Table 1, and the average correction factor shall be calculated from both ends of the roll at each calibration temperature and used for the entire length of the roll.

3.1.5.4 The roll shall not be used if the difference between the correction factors from each end of the sample sensors at any individual calibration temperature exceeds:

a. 1.0 °F or 0.6 °C for primary and secondary standard sensors.

b. 2.0 °F or 1.1 °C for control, recording, and load sensors, and SAT and TUS sensors.

3.1.5.5 The following shall apply to rolls not meeting the requirements of 3.1.5.4:

a. It is permitted to divide the roll into shorter lengths provided the shorter lengths meet the requirements of 3.1.5.4.

b. It is permitted to use individual sensors from the roll provided they are calibrated to the requirements of Table 1.

3.1.6 General Sensor Reuse Requirements

Reuse of any sensor is not permitted unless the insulation remains intact and the wire/cable including the measuring junction are not damaged (see Table 5).

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 5 - Sensor reuse and recalibration

Sensor Use	Sensor Type	Form	Recalibration[3]	Reuse[1][2][5]
Test Sensors TUS	Base and Refractory	Expendable	Recalibration of expendable base metal sensors is not permitted	• Quarterly when used at or below 500.0 °F or 260.0 °C • M, C, T, K, E: Limited by number of uses, temperature of use, and calendar days since first use (see 3.1.7.3) • J, N: Limited by number of uses, temperature of use, and calendar days since first use (see 3.1.7.3)
		Nonexpendable	M, C, T, J & N: Quarterly (see Table 1) E & K: Quarterly; permitted if used at or below 500.0 °F or 260.0 °C; not permitted if used above 500.0 °F or 260.0 °C	No other restrictions
	Noble[4]	Expendable	Semiannually (see Table 1)	No other restrictions
		Nonexpendable		
Test Sensor Nonresident SAT	Base and Refractory	Expendable	Recalibration of expendable base metal sensors is not permitted	• Quarterly when used at or below 500.0 °F or 260.0 °C • M, C, T, K, E: Limited by number of uses, temperature of use, and calendar days since first use (see 3.1.7.3) • J, N: Limited by number of uses, temperature of use, and calendar days since first use (see 3.1.7.3)
		Nonexpendable	M, C, T, J & N: Quarterly (see Table 1) E & K: Quarterly; permitted if used at or below 500.0 °F or 260.0 °C; not permitted if used above 500.0 °F or 260.0 °C	No other restrictions
	Noble[4]	Expendable	Semiannually (see Table 1)	No other restrictions

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Sensor Use	Sensor Type	Form	Recalibration[3]	Reuse[1][2][5]
		Nonexpendable		
Test Sensors Resident SAT	Base	Expendable	Recalibration of expendable base metal sensor is not permitted	M, E, J, K, T: Shall only be used at or below 500.0 °F or 260.0 °C N: Shall only be used at or below 1000.0 °F or 538.0 °C and is limited by the number of uses, temperature, and calendar days (see 3.1.7.3)
		Nonexpendable	M, T, J, N: Quarterly E & K: Quarterly; permitted if used at or below 500.0 °F or 260.0 °C (see Table 1)	M, E, J, K, T: Shall only be used at or below 500.0 °F or 260.0 °C N: No other restrictions
	Noble[4]	Expendable	Semiannually (see Table 1)	Shall only be used at or below 1000.0 °F or 538.0 °C
		Nonexpendable		No other restrictions
Control and Recording Sensors	All	All	Before first use	Replaced as stated in 2.4.42, 3.1.4.4, 3.4.8, and 3.4.9
Load Sensors	Base and Refractory	Expendable	Recalibration is not permitted	Limited by number of uses, temperature of use, and calendar days since first use (see 3.1.10.1, 3.4.8, and 3.4.9)
		Nonexpendable	M, C, T, J & N: Quarterly E & K: Quarterly; permitted if used at or below 500.0 °F or 260.0 °C; not permitted if used above 500.0 °F or 260.0 °C	Limited by number of uses, temperature of use, and calendar days since first use (see 3.1.10.2, 3.4.8, and 3.4.9)
	Noble[4]	Expendable	Semiannually except as stated in 3.4.8 and 3.4.9 (see Table 1)	No other restrictions except as stated in 3.4.8 and 3.4.9
		Nonexpendable		

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Sensor Use	Sensor Type	Form	Recalibration[3]	Reuse[1][2][5]

[1] For reuse of any Type E or Type K sensor used above 500.0 °F or 260.0 °C, the depth of insertion shall be equal to or greater than any previous use (see 3.1.7.1).
[2] See 3.1.6 and 3.1.7 for general reuse restrictions.
[3] Recalibration of any Type E or Type K sensor used above 500.0 °F or 260.0 °C is prohibited.
[4] Includes RTDs.
[5] Reuses: Example 1: Begin at room temperature, ramp to 900 °F or 480 °C for first TUS temperature, then ramp to 1200 °F or 650 °C for TUS max temperature, then cool back to room temperature. This is a single use.
Example 2: Begin at room temperature, ramp to 1325 °F or 718 °C for first production temperature, then cool to 1150 °F or 621 °C for the second production temperature, then cool back to room temperature. This is a single use.
Example 3: Begin at room temperature, ramp to 1900 °F or 1038 °C for first production temperature, then cool to 300 °F or 150 °C, then ramp up to 1100 °F or 593 °C for second production temperature, then cool back to room temperature. This is two uses even if the load remained closed inside the equipment.

3.1.7 SAT and TUS Sensor Reuse

3.1.7.1 During the reuse of Type E or Type K sensors when used above 500.0 °F or 260.0 °C, the depth of insertion shall be equal to or greater than any previous use.

3.1.7.2 Expendable base metal, noble metal, and refractory SAT and TUS sensors may be reused:

a. Base metal and refractory sensors may be used for up to 3 months from first use, without limit to number of uses, when used exclusively at or below 500.0 °F or 260.0 °C.

b. Noble metal sensors may be used for up to 6 months from first use, without limit to number of uses or temperature of use.

3.1.7.3 Expendable Base Metal and Refractory SAT and TUS Sensors

a. Types M, C, T, K, and E shall be limited to 3 months or five uses, whichever occurs first, between 500.0 °F and 1200.0 °F or between 260.0 °C and 650.0 °C and limited to a single use above 1200.0 °F or 650.0 °C.

b. Types J and N shall be limited to 3 months or ten uses, whichever occurs first, between 500.0 °F and 1200.0 °F or between 260.0 °C and 650.0 °C and limited to a single use above 1200.0 °F or 650.0 °C.

3.1.7.4 Records shall be maintained of the accumulated sensor reuse including sensor batch number, temperature, and use count.

Example 1: Sensors are replaced after a single use. Sensor batch number shall be recorded for the SAT and/or TUS.

Example 2: Sensors are reused. Sensor batch number, temperature, and use count shall be recorded for each use.

3.1.7.5 Base metal or refractory TUS sensors shall be limited to no more than the maximum number of uses defined in 3.1.7.3 for expendable sensors, 270 uses for nonexpendable sensors, or 6 months from first use, whichever occurs first, and may be reused subject to the limitations of 3.1.6 and 3.1.8 that are:

a. Used exclusively at ≤1200 °F or ≤650 °C.

b. Identified by the date of installation and by the number and temperatures of accumulated uses.

c. Preserved/protected from damage (i.e., crimping, excessive moisture contact, corrosion, etc.) between each TUS or remain installed on a rack that is protected between each TUS.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.1.8 Sensor Salvage

3.1.8.1 Salvage of damaged expendable sensors is permitted if the discrepant portion, including any portion previously exposed inside of the furnace, is removed and the hot junction remade.

3.1.8.2 The salvaged sensor's original calibration data shall be used.

3.1.9 Control and Recording Sensors

3.1.9.1 Control sensors shall be positioned in thermal processing equipment to ensure control and maintain temperature uniformity of the equipment within the qualified work zone. Recording sensors shall be located as stated in Table 9 with the applicable instrumentation type.

3.1.10 Base Metal Load Sensors

3.1.10.1 Expendable base metal and refractory load sensors may be used:

a. Up to 3 months after first use without limit to number of uses when used at or below 500.0 °F or 260.0 °C.

b. Types M, C, T, K, and E shall be limited to 3 months or five uses, whichever occurs first, between 500.0 °F and 1200.0 °F or between 260.0 °C and 650.0 °C and limited to a single use above 1200.0 °F or 650.0 °C.

c. Types J and N shall be limited to 3 months or ten uses, whichever occurs first, between 500.0 °F and 1200.0 °F or between 260.0 °C and 650.0 °C and limited to a single use above 1200.0 °F or 650.0 °C.

3.1.10.2 The life of nonexpendable base metal load sensors shall be limited by the maximum operating temperature and the number of calendar days since the first use.

3.1.10.3 Records shall be maintained of the accumulated load sensor use above 500 °F or 260 °C including sensor batch number, load cycle, temperature, and use count. The use count shall include uses during SAT and TUS. When sensors are replaced after a single use, the sensor batch number shall be recorded and traceable to the load.

3.1.10.4 The maximum replacement interval or number of uses, whichever occurs first since first use of nonexpendable base metal load sensors, shall comply with Table 6.

Table 6 - Nonexpendable base metal load sensor use temperature, interval or uses

Set Temperature		Maximum Replacement Interval or Number of Uses
>2300 °F	>1260 °C	1 use
>2200 to ≤2300 °F	>1205 to ≤1260 °C	Quarterly or 10 uses
>1800 to ≤2200 °F	>980 to ≤1205 °C	Quarterly or 90 uses
>1200 to ≤1800 °F	>650 to ≤980 °C	Quarterly or 180 uses
>500 to ≤1200 °F	>260 to ≤650 °C	Quarterly or 270 uses
≤500 °F	≤260 °C	Quarterly unlimited uses

3.1.10.5 When load sensors are used in multiple qualified temperature ranges, the shortest interval or use shall apply.

Example 1: A sensor with nine uses at 2250.0 °F or 1232.0 °C has only one more use allowed in the 2200.0 to 2299.9 °F or 1204.5 to 1259.9 °C range or any lower operating range. No uses remain at 2300.0 °F or 1260.0 °C or above.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Example 2: A sensor with 50 uses between 1400.0 and 1600.0 °F or 760.0 and 871.0 °C is then used at 1820.0 °F or 993.0 °C.

- The sensor has already exceeded the use limits for all ranges above 2199.9 °F or 1204.4 °C.
- The sensor is now subject to the 90-use limit as it has been used in a higher temperature range.

Example 3: A sensor with 50 uses between 1400.0 and 1600.0 °F or 760.0 and 871.0 °C is then used at 1015.0 °F or 546.0 °C.

- The sensor has already exceeded the use limits for all ranges above 2199.9 °F or 1204.4 °C.
- The sensor is subject to a 180-use limit as it has been used in the 1200.0 to 1799.9 °F or 648.9 to 982.2 °C temperature range.

3.1.11 Sensor Calibration Results and Records

3.1.11.1 The results of sensor calibration or recalibration shall be documented. The documentation shall include:

a. Identification of the sensor, batch of sensors, or wire/cable rolls.

b. Sensor type (e.g., K, N, E, RTD, etc.).

c. Date of calibration or recalibration.

d. Quantity or length of wire/cable rolls represented in calibration report.

e. Identification if the calibration was initial or a recalibration.

f. The required calibration accuracy.

g. Identification of the standard test sensor and standard test instrument used.

h. Nominal calibration temperatures.

i. Actual temperature readings of the sensor(s) being calibrated.

j. Calibration technique referencing ASTM E220 or other internationally recognized standards.

k. Correction factors or deviations/errors for each calibration temperature, including the average correction factor representing both ends for wire/cable rolls.

l. Documentation shall clearly state deviation (error) or correction factor.

m. A statement of traceability to NIST or other internationally recognized standards organization.

n. Identification of the calibration agency.

o. Identification of technician performing calibration.

p. Approval of an authorized agent for the calibration agency.

q. User quality organization approval.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.2 Instrumentation

3.2.1 General Instrumentation Requirements

3.2.1.1 Conversion from millivolts to degrees or degrees to millivolts shall be in accordance with ASTM E230 or other internationally recognized standards.

3.2.1.2 Output of sensors shall be converted to temperature readings by instruments specified herein or instruments of equal or greater accuracy.

3.2.1.3 Instruments shall be calibrated as stated in Table 7 and shall be traceable to NIST or other internationally recognized standards organization.

3.2.1.4 Process recording data collection shall be a minimum of six data points for each recorded sensor during each time at temperature processing cycle not to exceed 10-minute intervals.

3.2.1.4.1 For cycles where the time at temperature is <6 minutes, the data collection shall be ≤ once per minute.

3.2.1.5 Data collection intervals shall be sufficient to demonstrate conformance to soak time, heat-up, and cooling rate requirements, as applicable.

3.2.1.5.1 The recording instrument shall actively record/collect data during the entire time that parts or raw material are in the thermal processing equipment. In cases where load sensors must be disconnected to transfer the load, users shall have other objective evidence of process compliance (e.g., data showing temperature drop from the temperature of other furnace recording sensor(s) compliant with the soak period).

--`,`,,,,``,,`,`````,``,,``,,`-``,,`,,,,`-``,,`,,,,---

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 7 - Instruments and instrument calibration

Instrument	Instrument Type	Maximum Calibration Period[9]	Calibration Standard	Calibration Accuracy[1]	Use
Reference standard	Precision voltage reference	3 years	Precision Voltmeter or Voltage Reference Standard	NIST[6]	Limited to primary standard calibration[8]
Primary standard instrument	Potentiometer or digital voltmeter[1]	3 years	Reference standard	±0.1 °F or ±0.05 °C or ±0.015% of temperature reading, whichever is greater	Limited to laboratory calibration of secondary standard instruments, field test instruments, primary and secondary standard sensors[8]
Secondary standard instrument	Potentiometer or digital voltmeter[1]	Annually	Primary standard instrument	±0.3 °F or ±0.2 °C or ±0.05% of temperature reading, whichever is greater	Limited to laboratory calibration of field test instruments, SAT sensors, TUS sensors, load sensors, and control and recording sensors[8] (see 3.2.2.6)
Field test instrument[7]	SAT/TUS portable potentiometer or digital instrument, electronic data recorder, or data acquisition system	Quarterly	Primary or secondary standard instrument	±1.0 °F or ±0.6 °C or ±0.1% of temperature reading, whichever is greater	Limited to calibration of control or recording instruments performing SAT and TUS[5][8]
Control, over-temperature, recording, data acquisition instrument[7][11]	Digital instrument	Note[3]	Field test instrument	±2.0 °F or ±1.1 °C or 0.2% of temperature reading, whichever is greater[10]	Limited to control and recording the temperature of thermal processing equipment

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris

Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

(1) Instruments of equivalent or greater accuracy are acceptable.
(2) Applicable to quench or refrigeration equipment only.
(3) Instrument calibration intervals shall be:

Furnace Class	Control, over-temperature, recording, data acquisition instrument Interval	Over-temperature instruments used solely for furnace over-temperature protection Interval
1	Monthly	May be reduced to annually
2	Quarterly	
3	Quarterly	
4	Quarterly	
5	Semiannually	
6	Semiannually	
Refrigeration and quench equipment(4)	Semiannually	

(4) When the refrigeration and/or quench equipment sensor is connected to a channel in a furnace recording instrument, the interval for calibration of that channel in the recording instrument shall be the same calibration interval as the other channels in the instrument.
(5) Field test instruments meeting the accuracy requirements of secondary standards may be used to calibrate SAT and TUS instruments in the field. These instruments shall be calibrated quarterly using a primary standard.
(6) NIST or other internationally recognized standards organization. Per Manufacturer's Specifications for Reference Standard Instruments (e.g., stability of 2 µV/V per year) and NIST or equivalent Calibration for Reference Standard Instruments.
(7) Includes wireless transmission systems.
(8) Instruments shall be stored and used in the environment specified by the instrument manufacturer.
(9) Calibration periods for instruments shall begin on the date of calibration.
(10) Digital control instruments that only read in whole numbers shall have maximum calibration accuracy of ±2 °F or ±1 °C or ±0.2% of the temperature reading rounded inward toward the smaller whole number.
(11) For thermal processing equipment that has documented "out of use/service" periods beyond the due date and extension days in Table 22, calibration of process instrumentation shall be performed before being returned to service.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.2.2 Test Instruments (Primary Standard, Secondary Standard, and Field Test)

3.2.2.1 Primary and secondary standard instruments shall be digital and meet the calibration accuracy requirements of Table 7 in degrees of temperature or in millivolts that can be converted to demonstrate equivalent accuracy.

3.2.2.2 Field test instruments shall be digital and have a minimum readability of 0.1 °F or 0.1 °C for any input and output used.

3.2.2.3 Test instrument calibration shall be performed at a minimum of six simulated sensor input and/or output signals. These shall include the minimum and maximum of the operating range used for test or calibration and a minimum of four points in between either representing areas of normal operation or spaced at approximately equal intervals across the range in which the instrument is used for test or calibration.

3.2.2.3.1 For test instruments used only at a single temperature, the calibration shall be performed at that single temperature.

3.2.2.4 Calibration shall be performed for each type of input and output used, e.g., each sensor type in use and for mV, mA, etc., if the instrument is used for these scales.

3.2.2.5 Calibration shall be performed on each channel in use that can be altered or adjusted individually, or on each group of channels that can be altered or adjusted as a group. Channels not in use shall be blocked or tagged to prevent unintentional use.

3.2.2.6 A test instrument meeting the accuracy of a secondary standard may be used in a production environment as a field test instrument. The calibration frequency shall be the same as a field test instrument, and the instrument shall be operated within the environmental conditions specified by the instrument manufacturer.

3.2.3 Control, Recording, and Over-Temperature Instruments

3.2.3.1 All control, recording, and over-temperature instruments shall be digital.

3.2.3.2 Digital recording instruments shall produce permanent records with a minimum readability of 0.1 °F or 0.1 °C.

3.2.3.3 Instruments shall receive an unmodified signal from sensors, except for analog to digital and digital to analog conversions, or a digitally processed, error-checked equivalent representation of a direct measured value.

3.2.3.4 Calibration shall be performed in the as-found condition, taking into account any applied and documented offsets, at a minimum of three simulated sensor inputs (at the minimum, maximum, and at least one point in the middle third of the entire qualified operating temperature range) to document the as-found condition.

Example 1: A Class 2 ±10 °F furnace with a qualified operating temperature range of 300 to 1200 °F requires calibration at the Minimum = 300 °F, at the Maximum = 1200 °F, and at least 1 point in the middle third = 600 to 900 °F.

Example 2: A Class 2 ±6 °C furnace with a qualified operating temperature range of 200 to 1100 °C requires calibration at the Minimum = 200 °C, at the Maximum = 1100 °C, and at least 1 point in the middle third = 500 to 800 °C.

3.2.3.5 For equipment that does not have a defined qualified operating temperature range (e.g., quench tanks and refrigeration equipment), calibration shall be performed at a minimum of three simulated sensor inputs at the minimum, maximum, and at least one point in the middle third of the operating range used. Users shall have the ability to identify the operating range used for each piece of quench and/or refrigeration equipment.

3.2.3.5.1 Instrument calibration points for equipment with a qualified operating temperature range and an unqualified operating range of use, such as a cryogenic-temper unit, shall be as stated in 3.2.3.4 for the qualified operating range and as stated in 3.2.3.5 for the unqualified range of use.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.2.3.6 For control, recording, and over-temperature instruments used only at a single temperature, the instrument calibration shall either be performed at that single temperature of use or as stated in 3.2.3.4 or 3.2.3.5, as applicable.

3.2.3.7 The instrument error is the difference between the readings of the instrument being calibrated and the field test instrument. When any documented modification offsets are used in production, the reading of the instrument being calibrated shall be corrected algebraically.

3.2.3.8 Following any calibration adjustments, the instrument shall be verified as required in 3.2.3.4, 3.2.3.5, or 3.2.3.6, as applicable, to document the as-left condition.

3.2.3.9 Calibration shall be performed on each channel in use that can be altered or adjusted individually or on each group of channels that can be altered or adjusted as a group.

3.2.3.10 All active channels of multichannel digital recording instruments shall be calibrated. Channels not in use shall be identified to prevent unintentional use.

3.2.3.11 When the control and recording system is integrated such that the digitally displayed control value and the digitally recorded value are generated from the same measurement circuit and cannot be different, it is only necessary to document a single displayed/recorded value for the control reading.

3.2.3.12 For retort furnaces, the temperature of the furnace shall be controlled such that the specified temperature is maintained within the retort. As a minimum, the control instrument shall be calibrated across the temperature range of use within the retort.

3.2.3.13 Refrigeration and quench equipment control instruments that display temperature shall be calibrated. All recording instruments shall be calibrated (see Table 7).

3.2.3.14 The timing function for all digital recording instruments and data acquisition systems shall be calibrated at least annually and shall be accurate to ±1 min/h. The calibration may be performed for a time ≤1 hour and the results converted to meet ±1 min/h.

3.2.3.15 External timing devices, such as timers, clocks, stopwatches, etc., shall be calibrated at least every 2 years and shall be accurate to ±1 s/min.

3.2.3.16 As an alternative to 3.2.3.14 or 3.2.3.15, a defined digital time synchronization of digital recording instruments and data acquisition systems or external timing devices to NIST or other international equivalent via satellite, internet, or telephonic systems at least monthly to support a ±1 min/h accuracy for digital recording instruments and/or ±1 sec/min for external timing devices is acceptable.

3.2.3.17 Wireless equipment that performs the analog to digital conversion at the furnace and transmits a digital signal to the recording instrument is permitted. However, calibration of the complete wireless system (wireless transmitter, wireless receiver, and associated control and recording instruments) is required.

3.2.3.18 Correction Factors for Instruments

3.2.3.18.1 Extrapolation of calibration correction factors above the highest calibration temperature and below the lowest calibration temperature is prohibited by any calibration source except NIST or other internationally recognized standards organization.

3.2.3.18.2 Interpolation of correction factors between two known calibration points is permitted using the linear method. Alternatively, the correction factor of the nearest calibration point shall be used. Whichever method is used shall be defined and applied consistently.

3.2.4 Electronic Records

3.2.4.1 An electronic record is any combination of text, graphics, data, audio, pictorial, or other information represented in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.2.4.2 When using a system (control, recording, or data acquisition) that creates electronic records the system shall:

a. Create records that cannot be altered without detection.

b. Provide software and playback utilities as a means of examining and/or compiling the data but shall not allow the user any means for altering the source data.

c. Provide the ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and duplication.

d. Be capable of providing evidence the record was reviewed by recording an electronic review or a method of printing the record for a physical marking verifying review.

e. Support protection, retention, and retrieval of accurate records throughout the record retention period.

f. Ensure that the hardware and/or software shall operate throughout the retention period as specified in 3.7.

g. Provide methods of protection, such as a password, to limit system access to only individuals whose authorization is documented.

3.2.4.3 Evidence shall be provided that software revisions are verified to ensure continued compliance with the material or process specification requirements and, once installed, have not altered programs, recipes, or other means used to control thermal processing parameters.

3.2.5 Instrumentation Calibration Results and Records

3.2.5.1 The most recent successful calibration status shall be readily accessible at or in close proximity to the thermal processing equipment. As a minimum, the information shall include:

a. Instrument number or furnace number.

b. Date the calibration was performed.

c. Due date of the next calibration.

d. Identification of the technician who performed the calibration.

e. Indication of any limitations or restrictions of the calibration. A notation such as “see report” is acceptable.

3.2.5.2 The results of instrument calibration shall be documented. As a minimum, the documentation shall include:

a. Unique identification of the instrument.

b. Make and model of the instrument calibrated.

c. Unique identification of the test instrument used during calibration.

d. Identification of each sensor type in use (e.g., Type K, N, etc.) and form (e.g., V, mA, etc.) if the instrument is used for these scales.

e. Identification of location where signal was input (only required for measurement systems employing the alternate SAT).

f. Required calibration accuracy.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

g. As-found data at each calibration point and as-left data if any adjustments are made to include:

1. Nominal test temperature.

2. Reading of the instrument being calibrated.

3. Error or correction factor of test instrument (optional or when specified by the customer).

4. Error or correction factor of instrument under test (corrected or uncorrected for test instrument error when specified).

h. Correction and modification offsets as-found and as-left as stated in 3.2.6.

i Instrument calibration pass or fail statement.

j. Any limitations or restrictions of the calibration.

k. Date the calibration was performed.

l. Due date of the next calibration.

m. Statement of traceability to NIST or other internationally recognized standards organization.

n. Identification of the technician who performed the calibration.

o. Identification of the calibration agency if calibration is not performed internally.

p. Approval of an authorized agent for the calibration agency if not performed internally.

q. User quality organization approval.

3.2.6 Thermal Process Equipment Instrument Correction and Modification Offsets (see 2.4.38.1 and 2.4.38.2)

3.2.6.1 General Instrument Correction and Modification Offset Requirements

3.2.6.1.1 If instrument correction and/or modification offsets are used, a documented procedure shall exist describing when and how to make instrument correction and/or modification offsets.

3.2.6.1.2 The maximum cumulative correction offset shall not exceed the uniformity tolerance for the thermal processing equipment or ±10.0 °F or ±6.0 °C for refrigeration and quench instruments. (e.g., a Class 2 furnace instrument is limited to a maximum correction offset of ±10.0 °F or ±6.0 °C).

3.2.6.1.3 Instrument correction and/or modification offsets may be either internal (electronic) or external (manual) and shall be included in the calibration, SAT, and TUS documentation.

3.2.6.1.4 Controlling instrument modification offsets for TUS shall not exceed the allowances in Table 15 or Table 16.

3.2.6.1.5 TUS modification offsets are not permitted on recorder channels other than for the channel recording the control sensor temperature.

3.2.6.1.6 SAT modification offsets are not allowed. Control and recording instrument correction offsets for SAT shall not exceed the allowance in Table 15 or Table 16.

3.2.6.1.7 For load sensor systems used in production, correction offsets are allowed but not modification offsets.

3.2.6.1.8 When correction and/or modification offsets are implemented or changed, the effect of the new offset value over the entire qualified operating temperature range shall be evaluated. Objective evidence that the new offset value would not cause a failure of any other calibration, SAT, or TUS shall be documented.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.3 Thermal Processing Equipment

3.3.1 Furnace classes are defined in Table 8 and are based on the furnace class specified. When not specified, the furnace class shall meet the TUS requirements established in the governing specification for the parts or raw material being processed. Instrumentation types are defined by the level of instrumentation used to control, record, or indicate the desired temperature. Intervals for controlling and recording instrument calibration, SATs, and TUSs are based on the combined furnace class and instrumentation type stated in Table 7, Table 11, Table 12, Table 15, or Table 16.

Table 8 - Furnace class uniformity tolerances

Furnace Class	Temperature Uniformity Tolerance, °F	Temperature Uniformity Tolerance, °C(1)
1	±5.0	±3.0
2	±10.0	±6.0
3	±15.0	±8.0
4	±20.0	±10.0
5	±25.0	±14.0
6	±50.0	±28.0

(1) Some design authorities require TUS tolerances of ±5.0 °C and ±7.0 °C for Class 2 and Class 3 thermal processing equipment.

3.3.2 Requirements for furnace instrumentation type classification are shown in Table 9.

Table 9 - Instrumentation type requirements

Sensor(s) Required by Instrumentation Type	Instrumentation Type(1)					
	A	B	C	D+	D	E
One control sensor per zone that controls and displays temperature.	X	X	X	X	X	X
The temperature indicated by the control sensor in each control zone shall be recorded by a recording instrument. Alternatively, the recording instrument may be connected to a second sensor contained in the same sheath or holder as the control sensor and separated from the control sensor by no more than 0.38 inch or 10 mm.	X	X	X	X	X	
At least two additional recording sensors in each control zone shall be located to best represent the actual coldest and hottest temperatures in each control zone at any temperature of use based on the results from the most recent TUS. It is recognized that certain furnace designs/load configurations can prevent the location of these sensors in the precise coldest and hottest locations, but these sensors shall be located as close as practical (see 3.3.5 and 3.3.6). These recording locations may change over time. See 3.5.15 for relocation requirements.(2)	X		X			
At least one recording load sensor in each control zone. During production in multi-zone furnaces, empty zones (i.e., no material is placed in or intrudes into the zone) do not require a load sensor. However, a notation shall be made to the furnace load record that the zone was entirely empty.	X	X				
At least one additional recording sensor in each control zone located at least 3 inches or 76 mm from the control sensor position and of a different sensor type.(2)				X		
Each control zone shall have over-temperature protection (see 2.4.40).	X	X	X	X	X	

(1) Instrumentation types are listed in descending order of quality from left to right, i.e., A is better than B, etc.

(2) The over-temperature protection sensor may also be utilized as the recording sensor representing the hottest location for instrumentation Type A or C or as the additional recording sensor for Type D+ if in the proper location.

3.3.3 Instrumentation Requirements for Refrigeration Equipment and Quench Systems

3.3.3.1 All refrigeration equipment shall have a temperature controller. This temperature controller requirement is not applicable to liquid nitrogen, dry ice, or dry ice/liquid-cooled containers.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.3.3.2 All refrigeration equipment shall be equipped with a temperature recording instrument where time-at-temperature (minimum, maximum, or both) is required.

3.3.3.3 Unless otherwise specified, the requirements of 3.3.3.1 and 3.3.3.2 are not applicable during transportation of materials at sub-ambient temperatures.

3.3.3.4 Quench systems (immersion or spray) shall be equipped with a sensor that is recorded by recording instrument where temperature (minimum, maximum, or both) is required.

3.3.4 Additional Sensors

There is no limit to the number of additional recording or load sensors in any control zone, but their use shall be defined in controlled operating instructions or procedures.

3.3.5 Hot and cold temperature sensors, when required for instrument Types A and C, as stated in Table 10, may be inserted in heat sinks (see 2.4.27) and positioned at the hottest and coldest temperature locations of the control zone based on the most recent TUS when heat sinks of similar configuration are used during the TUS.

Table 10 - Minimum sensors required per control zone

Instrumentation Type	Minimum Number of Sensors Required				
	Hottest Temperature Sensors	Coldest Temperature Sensors	Load Sensors	Over-Temperature Sensor	Additional Sensor
A	1	1	1	1	N/A
B	N/A	N/A	1	1	N/A
C	1	1	N/A	1	N/A
D+	N/A	N/A	N/A	1	1
D	N/A	N/A	N/A	1	N/A

3.3.6 The minimum sensors required are shown in Table 10.

3.3.7 For multiple control zone furnaces with <225 ft^3 or <6.4 m^3 in total qualified work zone volume, it is acceptable to treat the furnace qualified work zone volume as a single control zone (see Table 10) for locating hot and cold temperature recording sensors (Type A or C instrumentation) and determining the number of over-temperature and load sensors required, regardless of the number of control sensors (Type A or B instrumentation), when the longest dimension of width, length, diameter, or height is no more than three times any of the other dimensions.

3.3.7.1 Multiple control zone furnaces with ≥225 ft^3 or ≥6.4 m^3 in total qualified work zone volume may be divided into control zones that shall not exceed 225 ft^3 or 6.4 m^3. Each control zone shall contain all the required sensors as defined in Table 10 for a single control zone. For furnaces that have individual control zones 225 ft^3 or 6.4 m^3 or greater in volume, each control zone shall contain all the required sensors as defined in Table 10.

3.4 System Accuracy Test

3.4.1 General SAT Requirements

3.4.1.1 The SAT is an assessment of the sum of the combined errors or corrections of the sensor, extension wire (and connectors), and instrument to ensure compliance with the requirements of Table 11 or Table 12.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 11 - Parts furnace class, instrument type, and SAT interval

Furnace Class	Temperature Uniformity Tolerance °F	Temperature Uniformity Tolerance °C	Minimum Instrumentation Type	Normal SAT Interval	Extended SAT Interval	Maximum Allowable SAT Difference, Whichever is Greater °F	Maximum Allowable SAT Difference, Whichever is Greater °C	Maximum Allowable SAT Difference, Whichever is Greater % of Reading
1	±5.0	±3.0	D, D+	Weekly	Weekly	±2.0	±1.1	±0.2
			B, C	Weekly	Biweekly			
			A	Biweekly	Monthly			
2	±10.0	±6.0	D, D+	Weekly	Weekly	±3.0	±1.7	±0.3
			B, C	Biweekly	Monthly			
			A	Monthly	Quarterly			
3	±15.0	±8.0	D	Biweekly	Monthly	±4.0	±2.2	±0.4
			D+	Biweekly	Bimonthly			
			B, C	Monthly	Quarterly			
			A	Quarterly	Semiannually			
4	±20.0	±10.0	D	Biweekly	Monthly	±4.0	±2.2	±0.4
			D+	Biweekly	Bimonthly			
			B, C	Monthly	Quarterly			
			A	Quarterly	Semiannually			
5	±25.0	±14.0	D	Biweekly	Monthly	±5.0	±2.8	±0.5
			D+	Biweekly	Bimonthly			
			B, C	Monthly	Quarterly			
			A	Quarterly	Semiannually			
6	±50.0	±28.0	D, E	Semiannually	Semiannually	±10.0	±5.6	±1.0
Refrigeration and quench equipment				Semiannually	Semiannually	±5.0	±2.8	N/A

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 12 - Raw material furnace class, instrument type, and SAT interval

Furnace Class	Temperature Uniformity Tolerance		Minimum Instrumentation Type	Normal SAT Interval	Extended SAT Interval	Maximum Allowable SAT Difference, Whichever is Greater		
	°F	°C				°F	°C	% of Reading
1	±5.0	±3.0	D, D+	Weekly	Monthly	±2.0	±1.1	±0.2
			B, C	Monthly	Quarterly			
			A	Monthly	Quarterly			
2	±10.0	±6.0	D, D+	Weekly	Monthly	±3.0	±1.7	±0.3
			B, C	Monthly	Quarterly			
			A	Monthly	Quarterly			
3	±15.0	±8.0	D	Biweekly	Monthly	±4.0	±2.2	±0.4
			D+	Biweekly	Bimonthly			
			B, C	Monthly	Quarterly			
			A	Quarterly	Semiannually			
4	±20.0	±10.0	D	Monthly	Quarterly	±4.0	±2.2	±0.4
			D+	Monthly	Four Months			
			B, C	Quarterly	Semiannually			
			A	Quarterly	Semiannually			
5	±25.0	±14.0	D	Monthly	Quarterly	±5.0	±2.8	±0.5
			D+	Monthly	Four Months			
			B, C	Quarterly	Semiannually			
			A	Quarterly	Semiannually			
6	±50.0	±28.0	D, E	Semiannually	Semiannually	±10.0	±5.6	±1.0
Refrigeration and quench equipment				Semiannually	Semiannually	±5.0	±2.8	N/A

3.4.1.2 The SAT shall be performed on all control and recording systems required by the applicable instrumentation type, as well as any additional recording systems used for parts and raw material acceptance in each control zone of each piece of thermal processing equipment used for production heat treatments.

3.4.1.2.1 Load sensors are not required to be in contact with parts, raw material, or any representation of parts or raw material during the SAT.

3.4.1.3 When the control and recording system is integrated such that the digitally displayed control value and the digitally recorded value are generated from the same measurement circuit and cannot be different, it is only necessary to document a single displayed/recorded value for the SAT.

3.4.1.4 The SAT shall be performed on additional systems used to justify SAT interval extension (see 3.4.3.2).

3.4.1.5 The SAT is not required for systems whose only function is over-temperature protection.

3.4.1.6 The SAT shall be performed using calibrated and independent SAT sensors meeting the requirements of Table 1 and calibrated and independent field test instruments meeting the requirements of Table 7.

3.4.1.7 Recording instruments used on thermal processing equipment shall not be used as a field test instrument unless it can be demonstrated that the SAT sensor recording channels of an integrated system are separated from the furnace recording system and also meet field test instrument requirements.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.4.2 Performing an SAT

3.4.2.1 The SAT shall be performed initially and periodically thereafter as stated in the interval requirements of Table 11 or Table 12.

3.4.2.2 For equipment that has documented “out of use/service” periods beyond the due date and extension days in Table 22, a new SAT shall be performed on all applicable systems before being returned to service.

3.4.2.3 An SAT (comparison, alternate, or reestablishing the relationship for the waiver) shall also be performed after any maintenance that could affect the SAT difference. Examples include, but are not limited to:

- Replacement of the sensor for the system being tested (i.e., systems utilizing individually calibrated sensors require an SAT upon replacement of each sensor, but systems utilizing sensors from the same calibrated roll/spool require an SAT upon replacement of the roll/spool).
- Replacement of the control or recording instrument.
- Calibration of the control or recording instrument when any adjustment has been made.
- If an internal correction offset is introduced or removed or an existing internal correction offset is altered.
- After implementation of corrective action due to a failed SAT.

3.4.2.4 The quality assurance organization shall make and document a determination whether equipment maintenance requires an SAT.

3.4.3 SAT Interval

3.4.3.1 The SAT interval shall be based upon the furnace class and instrumentation type (see 2.4.25).

3.4.3.1.1 For furnaces with multiple qualified operating temperature ranges, the SAT shall be performed using the most stringent interval for the furnace class used.

3.4.3.2 The SAT interval may be extended to the maximum allowed SAT interval in Table 11 or Table 12, if a preventive maintenance program is in effect as described in 2.4.42, and one of the following conditions is met:

3.4.3.2.1 At least two sensors in each control zone are nonexpendable Type B, N, R, or S.

3.4.3.2.2 The relationship requirements of 3.4.9.6, 3.4.9.6.2, and Table 14 are documented and met.

3.4.3.2.2.1 If the weekly relationship exceeds 2.0 °F or 1.1 °C, a comparison SAT shall be performed prior to returning the thermal processing equipment to production and the interval shall return to the normal interval until the next TUS is performed to establish the new relationship.

3.4.4 SAT Difference

3.4.4.1 The maximum allowable SAT difference shall be defined by the furnace class as shown in Table 11 or Table 12.

3.4.4.2 The difference calculated between the readings of the sensor system being tested (instrument/extension wire/sensor) and the corrected reading of the SAT sensor and SAT instrument (after the SAT sensor and SAT instrument correction factors have been applied algebraically) shall be documented as the SAT difference.

3.4.4.2.1 The correction factors for the SAT sensor and SAT instrument may be determined by the SAT set temperature or the observed SAT instrument reading.

3.4.4.2.2 Whichever method is used shall be defined and applied consistently.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.4.4.3 The temperature readings from the process instrument and sensor being compared with the SAT sensor and field test instrument shall be the temperature reading, read or recorded during production heat treatment. Certain offsets, if consistently applied during production heat treatment as stated in 3.4.4.4 and supported by documented procedures, shall be algebraically applied to the system being tested.

3.4.4.4 Allowable instrument correction or modification offsets include:

3.4.4.4.1 Internal modification offsets applied to the control and control recording channel instruments solely to correct a skewed TUS result.

Example: Reading from control instrument is 1000.0 °F and there is a -3.0 °F modification offset applied electronically to the control instrument. Then +3.0 °F shall be added to the 1000.0 °F reading before calculating the difference in 3.4.4.2.

3.4.4.4.2 A previously documented and specified manual correction offset to the control or recording instrument to correct an SAT difference.

Example: Reading from control instrument is 1352.0 °F and there is a +2.0 °F correction offset for SAT applied manually. Then -2.0 °F shall be added to the 1352.0 °F control reading before calculating the difference in 3.4.4.2.

3.4.5 Prohibited instrument offsets include:

3.4.5.1 External (manual) modification offset applied to the control instrument that has been specified for production solely to correct a skewed TUS result. These manual modification offsets have no effect on the performance of the SAT or calculation of the SAT difference.

Example: Reading from control instrument is 1905.0 °F and there is a +5.0 °F modification offset for TUS applied manually. Then the 1905.0 °F control reading is used when calculating the difference in 3.4.4.2.

3.4.6 SAT Methods

3.4.6.1 The SAT requirement may be fulfilled using any one of three methods interchangeably, provided the sensor system meets the requirements of the method implemented:

3.4.6.1.1 Comparison SAT (see 3.4.7).

3.4.6.1.2 Alternate SAT (see 3.4.8).

3.4.6.1.3 SAT waiver (see 3.4.9).

--`,,,,`,,`,`,,`,`,,`,,`,,`-`-`,,`,,`,`,,,,-`-`,,`,,,,---

3.4.7 Comparison SAT

3.4.7.1 The displayed temperature indication and recording of the sensor being tested as used in production, with appropriate offsets or correction factors, at any temperature within the qualified operating temperature range(s), shall be compared with the corrected temperature indication of the SAT sensor on a test instrument.

3.4.7.1.1 For furnaces with multiple qualified operating temperature ranges, a periodic SAT shall be performed in each range at least annually.

3.4.7.2 The tip (measuring junction) of the SAT sensor shall be as close as practical to the tip (measuring junction) of the control, or recording, sensor, but the sensor tip to tip distance shall not exceed 3 inches or 76 mm.

3.4.7.3 The SAT sensor may be inserted temporarily to perform the comparison SAT or may be a resident SAT sensor subject to the following limitations:

3.4.7.3.1 The resident SAT sensors shall be restricted to Type B, R, S, or N at temperatures greater than 500 °F or 260 °C and shall be nonexpendable at temperatures greater than 1000 °F or 538 °C (see Table 5).

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.4.7.3.2 The resident SAT sensor type shall be different from that of the sensor being tested as defined by Table 13.

Table 13 - Allowable combinations of resident SAT sensor and sensor being tested for temperatures greater than 500 °F or 260 °C

Resident SAT Sensor Type	Control or Recording Sensor Being Tested				
	B	R	S	N	All Other Sensor Types
B	No	Yes	Yes	Yes	Yes
R	Yes	No	No	Yes	Yes
S	Yes	No	No	Yes	Yes
N	Yes	Yes	Yes	No	Yes

3.4.7.3.3 The resident SAT sensor shall be affixed in position to an assembly to prevent movement in relation to the sensor under test. The position shall be verified on installation and replacement. Alternatively, the resident SAT sensor may be located independently of the sensor under test and its position shall be verified before each SAT to ensure that it has not moved between sensor installation events.

3.4.7.3.4 It is permitted to use the over-temperature sensor as the resident SAT sensor provided it is used only for over-temperature protection and meets all of the requirements of an SAT sensor, 3.4.7.3, and Table 13.

3.4.8 Alternate SAT

3.4.8.1 The alternate SAT method shall only be applied to the following sensor systems:

3.4.8.1.1 Load sensor systems (expendable and nonexpendable) where the sensor is used only once (single use) at temperatures >500 °F or >260 °C.

3.4.8.1.2 Load sensor systems (expendable and nonexpendable) used at temperatures ≤500 °F or ≤260 °C where the sensor is replaced at the same or less than the SAT interval.

3.4.8.1.3 Control and recording sensors (e.g., hot and/or cold recording sensors for instrumentation Types A or C and the additional recording sensor for instrumentation Type D+) where the sensor is replaced at the same or less than the SAT interval. Section 3.1.10 shall also apply to base metal recording sensors when the alternate SAT is used.

3.4.8.2 Alternate SAT Method

Periodic calibration of control and/or recording instruments as stated in 3.2.3.4 or 3.2.3.5, as applicable, and meeting the requirements of Table 7 shall be performed from the point at which the sensor will be connected (including the instrument/extension wire/connector) and one of the following three options are met:

3.4.8.2.1 The sum of the sensor calibration error plus the instrument calibration error or sensor calibration correction factor plus the instrument calibration correction factor shall meet the maximum SAT difference requirements of Table 11 or Table 12, as appropriate.

3.4.8.2.2 Use the appropriate sensor and instrument correction factors applied manually or via programming, as allowed by the calibration limits of Table 1 and Table 7, so that data used from the control and recording instrument combined meets the maximum SAT difference requirements of Table 11 or Table 12, as appropriate.

3.4.8.2.3 Limit instrument calibration error or correction factor and/or sensor error or correction factor such that the sum of the errors or correction factors cannot exceed the maximum SAT difference defined by Table 11 or Table 12, accordingly. The sensor lot identification used in the equipment during the alternate SAT period shall be documented.

Example 1: Maximum SAT difference for a Class 5 furnace is the greater of ±5.0 °F or ±3.0 °C or ±0.5% (see Table 11). Limiting instrument calibration to ±1.0 °F or ±.0.6 °C or ±0.1% and limiting sensor calibration to ±2.0 °F or ±1.1 °C or ±0.3% will always meet the maximum SAT difference.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Example 2: Maximum SAT difference for a Class 2 furnace is the greater of ±3.0 °F or ±1.7 °C or ±0.3% (see Table 11). Limiting instrument calibration to ±1.0 °F or ±0.6 °C or ±0.1% and limiting sensor calibration to ±2.0 °F or ±1.1 °C or ±0.2% will always meet the maximum SAT difference.

3.4.8.3 Alternate SAT Frequency

The alternate SAT method shall be performed and documented when any of the following occurs:

- Systems utilizing individually calibrated sensors require the alternate SAT upon replacement of each sensor.
- Systems utilizing sensors from the same calibrated roll/spool require an alternate SAT upon replacement of the roll/spool.
- Calibration of the system instrument (control or recording).

3.4.8.4 The alternate SAT calculation shall be performed at each temperature on the most recent instrumentation calibration report using sensor data from the sensor calibration report applicable to the sensor used. Sections 3.1.4.8 through 3.1.4.10 shall apply for sensor temperatures.

3.4.9 SAT Waiver

The SAT waiver requirements are as follows:

3.4.9.1 There shall be at least two recording load sensors in each control zone.

3.4.9.2 Noble metal load sensors shall be either replaced or recalibrated quarterly.

3.4.9.3 Base metal load sensors shall be controlled as follows:

3.4.9.3.1 Expendable base metal load sensors shall be single use only.

3.4.9.3.2 Nonexpendable base metal load sensors shall meet the replacement requirements in Table 6 and shall be recalibrated or replaced anytime that observations, made and recorded at least weekly, reveal any unexplainable difference between their readings and the readings of the two recording sensors in 3.4.9.6.

3.4.9.4 Noble metal control and recording sensors shall be replaced at least every 2 years.

3.4.9.5 Base metal control and recording sensors shall be replaced at least quarterly.

3.4.9.6 Weekly readings at one thermal process set point measured within 5 minutes of the end of a thermal process soak period shall be compared. The relationship between the control sensor and an additional sensor in each control zone shall remain within 2.0 °F or 1.1 °C of their relationship determined at the time of the most recent TUS (at the nearest temperature tested during the most recent TUS). The additional sensor may be the over-temperature sensor. Weekly readings shall be taken from the same sensor pair between two consecutive TUS.

3.4.9.6.1 The two sensors chosen shall be different types following the same restrictions as resident SAT sensors in Table 13. The additional recording sensor shall be restricted to Type B, R, S, or N at temperatures greater than 500 °F or 260 °C and shall be nonexpendable at temperatures greater than 1000 °F or 538 °C (see Table 5).

3.4.9.6.2 During each periodic TUS, the relationship between the control sensor system and the additional sensor system shall be determined by calculating the temperature of the control sensor minus the temperature of the additional sensor within the final 5 minutes (single reading or average of the last 5 minutes) of the TUS soak period to determine the relationship at each TUS temperature (see Table 14).

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.4.9.6.3 If the weekly relationship exceeds 2.0 °F or 1.1 °C, then one of the following options shall apply:

3.4.9.6.3.1 A successful SAT using one of the other methods (comparison or alternate) shall be performed before returning to production.

3.4.9.6.3.2 After 4.5, 4.6, 4.7, and 4.8 are completed, the SAT waiver relationship shall be reestablished by performing a TUS at all required temperatures of the initial TUS. The TUS interval shall continue at the interval in use at the time of the last periodic TUS.

3.4.10 SAT Difference Pass/Fail Requirements

3.4.10.1 It is not permitted to apply modification offsets to achieve an acceptable SAT.

3.4.10.2 If the calculated SAT difference exceeds the allowable difference of Table 11 or Table 12, the failure shall be documented, the cause of the difference determined, and corrective action taken before commencing additional thermal processing. The requirements of Section 4 shall apply.

3.4.10.3 If the cause is wholly or partially a result of movement of the sensor being tested from its documented location and depth of insertion, it shall be returned to its original documented location and depth of insertion and the SAT repeated.

3.4.10.4 Instrument recalibration or adjustment, including any correction offset of the control or recording instrument calibration, is permitted within the maximum limitations of Table 15 or Table 16. The effect of this recalibration or adjustment over the entire qualified operating temperature range shall be evaluated and objective evidence that the correction offset would not cause a failure of any other calculated SAT or TUS shall be documented. The requirements of 3.4.2.3 and Section 4 shall apply.

3.4.10.5 All corrective actions shall be documented. After corrective action has been implemented and prior to any additional thermal processing, the SAT shall be repeated and yield an acceptable SAT difference. The results of both the failed and acceptable SAT shall be documented.

3.4.11 SAT Results and Records

3.4.11.1 Comparison SAT (see 3.4.7)

The results of the comparison SAT shall be documented. At a minimum, the documentation for each sensor system tested shall include:

a. Identification of the sensor system being tested.

b. Identification of the SAT sensor and depth of insertion when Type K or Type E is reused.

c. Identification of the SAT instrument.

d. Date and time of the SAT.

e. Set point of the thermal processing equipment during the SAT. The test temperature shall be documented as the SAT set point for thermal processing equipment that does not have a set point temperature.

f. Observed control or recording instrument readings and recordings.

g. Observed SAT instrument readings.

h. SAT sensor correction factor.

i. SAT instrument correction factor.

j. Corrected SAT instrument reading.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

k. Calculated SAT difference (control and recording instrument readings minus the corrected SAT instrument reading).

l. As-found and as-left correction and/or modification offsets if used during production.

m. SAT difference pass or fail statement.

n. Identification of the technician who performed the SAT.

o. Identification of the agency if SAT is not performed internally.

p. Approval of an authorized agent for the calibration agency if not performed internally.

q. User quality organization approval.

3.4.11.2 Alternate SAT (see 3.4.8)

The results of the alternate SAT shall be documented. At a minimum, the documentation for each sensor system tested shall include:

a. Identification of the sensor system being tested.

b. Identification of the sensor or roll.

c. Date and time of the alternate SAT.

d. Method used:

1. Sum of calibration errors or correction factors (see 3.4.8.2.1).

2. Correction factors applied manually or via programming (see 3.4.8.2.2).

3. Limit instrument/thermocouple calibration errors or correction factors such that SAT is always met (see 3.4.8.2.3).

e. Final SAT results calculated for the method used with supporting documentation.

f. Identification of the technician who performed the alternate SAT, as applicable.

g. Identification of the agency if the alternate SAT is not performed internally.

h. Approval of an authorized agent for the agency performing the alternate SAT if not performed internally.

i. User quality organization approval.

3.4.11.3 SAT Waiver (see 3.4.9)

--`,`,,,,``,,`,`````,``,,``,,`-```,,`,,,,-``,,`,,,,---

The results of the SAT waiver shall be documented. At a minimum, the documentation for each piece of thermal processing equipment shall include:

a. Identification of the thermal processing equipment using the SAT waiver.

b. Identification of the control and additional sensor system used for the relationship test.

c. Date of installation of the control and additional sensor used for the relationship test.

d. Date and temperature(s) of the most recent TUS.

e. Documentation of the relationship at each TUS temperature (see Table 14).

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

f. Documentation of the weekly readings and relationship (see Table 14).

g. Identification of the agency if the SAT waiver is not performed internally.

h. Approval of an authorized agent for the agency performing the SAT waiver if not performed internally.

i. User quality organization approval.

Table 14 - SAT waiver relationship

	Calculation	Examples		
Reference relationship established during TUS (Rref)	Control sensor average = $\bar{C} = \frac{T°_{ctrl\,1} + T°_{ctrl\,2} + \ \ldots \ + T°_{ctrl\,n}}{n}$	1000.4 °F 538.0 °C		
	Additional sensor average = $\bar{A} = \frac{T°_{add\,1} + T°_{add\,2} + \ \ldots \ + T°_{add\,n}}{n}$	1001.2 °F 538.4 °C		
	Relationship (Rref) = $R = \ \bar{C} - \bar{A}$	-0.8 °F -0.4 °C		
Weekly Readings Examples				
Relationship established during weekly readings (Rw)		$Week_n$	$Week_{n+1}$	$Week_{n+2}$
	Control sensor reading $T°_{ctrl}$	1000.0 °F 537.8 °C	1000.4 °F 538.0 °C	1000.4 °F 538.0 °C
	Additional sensor reading $T°_{add}$	1001.3 °F 538.5 °C	999.6 °F 537.6 °C	998.3 °F 536.8 °C
	Weekly relationship (Rw) = $T°_{ctrl}$ - $T°_{add}$	-1.3 °F -0.7 °C	+0.8 °F +0.4 °C	+2.1 °F +1.2 °C
	Weekly comparison (Rw) - (Rref)	-0.5 °F -0.3 °C	+1.6 °F +0.8 °C	+2.9 °F +1.6 °C
	Result pass/fail	Pass	Pass	Fail

3.5 Temperature Uniformity Surveys (TUS)

3.5.1 General TUS Requirements

3.5.1.1 The TUS shall be performed using calibrated and independent TUS sensors meeting the requirements of Table 1 and calibrated and independent TUS instruments meeting the requirements of Table 7.

3.5.1.2 Recording instruments used on thermal processing equipment shall not be used to record TUS sensor temperatures unless it can be demonstrated that the TUS sensor recording channels of an integrated system are separated from the thermal processing equipment recording system and also meet field test instrument requirements.

3.5.1.3 An initial TUS shall be performed to determine the temperature uniformity and establish the qualified work zone(s) and qualified operating temperature range(s).

3.5.1.4 During initial and periodic TUS, the dimensions of the measuring junction locations of the corner sensors for a square or rectangle work zone or the periphery sensors of a cylindrical work zone shall define the extremities of the work zone such that no material heat treated extends beyond these boundaries.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 15 - Parts furnace class, instrument type, and TUS interval

Furnace Class	Temperature Uniformity Tolerance		Minimum Instrument Type	Normal Periodic TUS Interval	Required Number of Consecutive Successful Periodic TUS	Extended Periodic TUS Interval	Maximum Permitted Offset(1)(2)(3)		
	°F	°C					°F	°C	Percent of Maximum Qualified Operating Temperature
1	±5.0	±3.0	D, D+	Monthly	8	Bimonthly	±2.5	±1.5	N/A
			B, C	Monthly	4	Quarterly			
			A	Monthly	2	Semiannually			
2	±10.0	±6.0	D, D+	Monthly	8	Bimonthly	±5.0	±3.0	N/A
			B, C	Monthly	4	Quarterly			
			A	Monthly	2	Semiannually			
3	±15.0	±8.0	D, D+	Quarterly	4	Semiannually	±8.0	±5.0	0.38
			B, C	Quarterly	3	Semiannually			
			A	Quarterly	2	Annually			
4	±20.0	±10.0	D, D+	Quarterly	4	Semiannually	±10.0	±6.0	0.38
			B, C	Quarterly	3	Semiannually			
			A	Quarterly	2	Annually			
5	±25.0	±14.0	D, D+	Quarterly	4	Semiannually	±13.0	±7.0	0.38
			B, C	Quarterly	3	Semiannually			
			A	Quarterly	2	Annually			
6	±50.0	±28.0	D, E	Annually	None	Annually	N/A	N/A	0.75
Unless otherwise specified, a TUS of refrigeration and quench equipment is not required.							±10.0	±6.0	N/A

(1) The maximum permitted offset shall be the same for manual and electronic methods.
(2) °F or °C or % of reading, whichever is greater.
(3) The maximum is permitted for TUS modification offsets and SAT correction offsets separately.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 16 - Raw material furnace class, instrument type, and TUS interval

Furnace Class	Temperature Uniformity Tolerance		Minimum Instrument Type	Normal Periodic TUS Interval	Required Number of Consecutive Successful Periodic TUS	Extended Periodic TUS Interval	Maximum Permitted Offset(1)(2)(3)		
	°F	°C					°F	°C	Percent of Maximum Qualified Operating Temperature
1	±5.0	±3.0	D, D+	Monthly	8	Semiannually	±2.5	±1.5	N/A
			B, C	Quarterly	4	Semiannually			
			A	Quarterly	2	Semiannually			
2	±10.0	±6.0	D, D+	Monthly	8	Semiannually	±5.0	±3.0	N/A
			B, C	Quarterly	4	Semiannually			
			A	Quarterly	2	Semiannually			
3	±15.0	±8.0	D, D+	Quarterly	4	Semiannually	±8.0	±5.0	0.38
			B, C	Semiannually	3	Annually			
			A	Semiannually	2	Annually			
4	±20.0	±10.0	D, D+	Quarterly	4	Semiannually	±10.0	±6.0	0.38
			B, C	Semiannually	3	Annually			
			A	Semiannually	2	Annually			
5	±25.0	±14.0	D, D+	Quarterly	4	Semiannually	±13.0	±7.0	0.38
			B, C	Semiannually	3	Annually			
			A	Semiannually	2	Annually			
6	±50.0	±28.0	D, E	Annually	None	Annually	N/A	N/A	0.75
Unless otherwise specified, a TUS of refrigeration and quench equipment is not required.							±10.0	±6.0	N/A

(1) The maximum permitted offset shall be the same for manual and electronic methods.
(2) °F or °C or % of reading, whichever is greater.
(3) The maximum is permitted for TUS modification offsets and SAT correction offsets separately.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.2 Initial TUS Temperatures

3.5.2.1 The initial TUS shall be performed at the minimum and maximum temperatures of each qualified operating temperature range(s).

3.5.2.2 Additional TUS temperatures shall be added to ensure that there are no more than 600 °F or 335 °C increments between adjacent TUS temperatures.

Example: If a furnace is used from 800 to 1800 °F or 425 to 980 °C, the furnace shall be surveyed at 800 °F or 425 °C, 1800 °F or 980 °C, and one intermediate temperature to meet the maximum permitted 600 °F or 335 °C range requirement. Performing a TUS at any temperature from 1200 to 1400 °F or 645 to 760 °C would satisfy the 600 °F or 335 °C range requirement.

3.5.2.3 It is not required to include hottest and coldest recording sensors for instrument Type A and C equipment during the initial TUS. The initial TUS defines the location of these sensors.

3.5.2.4 For thermal processing equipment used only at a single set point temperature, the TUS shall be performed at that single temperature.

3.5.3 Periodic TUS Temperatures

3.5.3.1 Periodic TUS shall be performed at any temperature within each qualified operating temperature range(s).

3.5.3.2 A periodic TUS shall be performed at the minimum of each qualified temperature range and at the maximum temperature of each qualified operating temperature range at least annually.

3.5.3.3 TUS Temperatures for Equipment with Single Qualified Operating Temperature Range

3.5.3.3.1 For single operating temperature ranges greater than 600 °F or 335 °C, during each periodic TUS, temperatures shall be selected so that one temperature is within 300 °F or 170 °C of the maximum and another temperature is within 300 °F or 170 °C of the minimum of qualified operating temperature range(s) and there are no more than 600 °F or 335 °C increments in between adjacent TUS temperatures.

Example: If the qualified operating temperature range is 200 to 1200 °F ±10 °F or 90 to 650 °C ± 6 °C, a TUS temperature shall be selected between 200 and 500 °F or 90 and 260 °C and between 900 and 1200 °F or 480 and 650 °C. No two adjacent TUS temperatures shall be greater than 600 °F or 335 °C apart.

A TUS at 350 °F or 177 °C and at 950 °F or 510 °C would be acceptable. However, a TUS at 250 °F or 121 °C and 1000 °F or 538 °C would not be acceptable since these temperatures exceed the 600 °F or 335 °C separation increment.

3.5.3.4 TUS Temperatures for Equipment with Multiple Qualified Operating Temperature Ranges

3.5.3.4.1 For equipment with multiple operating temperature ranges, each sub-range shall be treated as a single operating temperature range and the requirements of 3.5.3.1, 3.5.3.2, and 3.5.3.3 apply to each sub-range.

Example 1: A furnace may be qualified to operate from 600 to 1000 °F ± 10.0 °F or 315 to 540 °C ± 6.0 °C and from 1000 to 1800 °F ± 25.0 °F or 540 to 980 °C ± 14.0 °C. The furnace contains two separate qualified operating temperature ranges.

A furnace meeting ±10.0 °F at 1000 °F or ±6.0 °C at 540 °C automatically meets ±25.0 °F at 1000 °F or ±14.0 °C at 540 °C; therefore, a duplicate TUS at 1000 °F or 540 °C is not required.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Example 2: If the qualified operating temperature ranges are 800 to 1025 °F ± 10.0 °F or 425 to 550 °C ± 6.0 °C, 1025 to 1400 °F ± 15.0 °F or 550 to 760 °C ± 8.0 °C and 1400 to 1600 °F ± 25.0 °F or 760 to 870 °C ± 14.0 °C.

Annually, the TUS would be performed at: 800 °F and 1025 °F or 425 °C and 550 °C and meet ±10.0 °F or ±6.0 °C, 1400 °F or 760 °C and meet ±15.0 °F or ±8.0 °C, and 1600 °F or 870 °C and meet ±25.0 °F or ±14.0 °C.

3.5.4 Equipment Modifications and Repairs

When maintenance is performed on thermal processing equipment, the maintenance task shall be documented, and a determination shall be made and documented by the user quality assurance organization whether any testing is required before returning the equipment into service. This determination may require a new initial TUS, performing an additional TUS, or no testing at all.

3.5.4.1 Major Modifications and Repairs

3.5.4.1.1 A new initial TUS and SAT shall be performed after any of the following equipment modifications, repairs, or adjustments that could have altered the thermal characteristics of the equipment. Examples where an initial TUS and SAT is required include, but are not limited to, the following:

a. Relocation of thermal processing equipment. The initial TUS may be waived if the thermal processing equipment is designed to be portable (i.e., the thermal processing equipment has permanent wheels or other means of portability), but in some cases a new periodic TUS can be necessary.

b. Increase in the maximum qualified operating temperature or decrease in the minimum qualified operating temperature.

c. Burner size, number, type, or location change.

d. Heating element number, type, or location change.

e. Changes to airflow pattern/velocity such as baffle positions, fan speed, fan quantity, etc.

f. Change of refractory thickness.

g. New refractory with different thermal properties.

h. Change of vacuum furnace hot zone design or materials.

i. Change of control sensor (e.g., type, thickness of sensor assembly, gauge of the sensor elements, or hot junction construction).

j. Change of the control sensor location.

k. Change of combustion pressure settings from their original settings.

l. Change of furnace operating atmosphere damper system settings from their original settings.

m. Control instrument or program change:

    1. Proportional versus high-low/off-on.

    2. Change of the control instrument model or type.

    3. PLC logic program change to the furnace heat control scheme.

    4. Adjustment of control instrument tuning constants, parameters, or rheostats.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

n. Qualified work zone volume increase covering a volume not previously surveyed.

o. Qualified work zone location change covering a volume not previously surveyed.

3.5.5 Periodic TUS Intervals

3.5.5.1 A periodic TUS shall be performed at the normal periodic TUS interval as stated in Table 15 or Table 16 based on furnace class and instrumentation type.

3.5.5.2 Extended periodic TUS intervals may be used based on furnace class, instrument type, and history of the required number of consecutive successful periodic TUS shown in Tables 15 and 16 after an initial TUS. In addition, a documented equipment preventive maintenance program, as defined in 2.4.42, shall be in effect.

3.5.5.3 If equipment modifications, repairs, or adjustments as described in 3.5.4.1.1 are made, the TUS interval shall revert to the normal periodic TUS interval until the required number of consecutive successful periodic TUS are completed.

3.5.6 Equipment Parameters During the TUS

3.5.6.1 During each TUS, except as outlined in 3.5.7 and 3.5.8, all parameters shall reflect the normal operation of the equipment during production. Examples of normal equipment operation include, but are not limited to, the following:

a. If the doors of a continuous furnace are normally open during production, they shall also be open during the TUS.

b. If slow heat-up rates and stabilization temperatures are not used during production, they shall not be used during the TUS.

--`,,,,``,,`,`````,``,,``,,`-``,,`,,,,-``,`,,,,---

c. Any documented ramp rate used during production shall be acceptable when performing the TUS.

d. If excess combustion air is used during production, it shall also be used during the TUS.

e. If circulating fans are operated during production, they shall also be operated during the TUS.

3.5.7 Equipment Temperature When TUS Sensors are Inserted

3.5.7.1 If the normal operation of the equipment when used during production is to load parts or raw material into a hot furnace, it is acceptable to insert the TUS sensors into the furnace with the furnace cold or with the furnace stabilized at or below the TUS temperature.

3.5.7.2 If the normal operation of the equipment when used during production is to load parts or raw material into a cold furnace, preheating the furnace prior to the coldest TUS temperature to perform the TUS is not permitted. Ramping from the coldest TUS temperature to a higher test temperature is permitted.

3.5.8 Load Condition and Atmospheres During the TUS

3.5.8.1 A TUS may be performed with a production load, simulated production load, a rack, or empty. Once a method of surveying the equipment is established during the initial TUS, subsequent periodic TUS shall be conducted using the same method. If changes are made to the established method, an initial TUS shall be performed to validate the revised method and the TUS interval shall revert to the normal periodic TUS interval until the required number of consecutive successful periodic TUS are completed.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.8.2 If the TUS is performed empty or with a rack, and if TUS sensors are attached to or inserted into heat sinks, the side-to-side thickness or diameter of the heat sink shall not exceed 0.5 inch or 13 mm and shall not exceed the thickness of the thinnest material being processed in that equipment. Heat sink material shall be the material with the highest room-temperature thermal conductivity consistent with the predominant part or raw material processed in the equipment.

3.5.8.3 When the TUS is performed with a load and the TUS sensors are attached to simulated parts or raw material, the load shall represent the thickness of the material normally used during production.

3.5.8.4 The atmosphere used during a TUS shall be the normal atmosphere used during production. Equipment used for processes whose required atmospheres could contaminate the TUS sensors (i.e., carburizing, nitriding, and endothermic and exothermic gases) or atmospheres that could pose a safety hazard (i.e., hydrogen- or ammonia-containing gases) may be surveyed with an atmosphere of air or inert gas.

3.5.8.5 The furnace vacuum level used during the TUS shall be the lowest vacuum level used during production but need not be less than 1.0×10^{-3} mm Hg (1×10^{-3} Torr or 1.3×10^{-3} millibar).

3.5.8.6 For vacuum furnaces operated in partial pressure (with backfill gas) during production, at least one of the periodic TUSs shall be performed annually at a minimum of one operating temperature and within the partial pressure range used in production. The gas used shall be one of the gases used during production.

3.5.9 TUS Requirements for Batch Furnaces, Salt Baths, Controlled Temperature Liquid Baths, and Fluidized Bed Furnaces (Excluding Controlled Temperature Quench Baths)

3.5.9.1 The number of TUS sensors required during the TUS shall be as stated in Figure 1 and Table 17 and Table 19. An initial TUS of multiple control zone furnaces with Type A or C instrumentation shall be performed with sufficient additional TUS sensors to adequately evaluate the temperature extremes of each control zone to identify hot and cold recording sensor locations.

Table 17 - Minimum number of TUS sensors and required locations

Qualified Work Zone Volume		Minimum Number of TUS Sensors	Qualified Work Zone Shape	Required Location of TUS Sensors[(1)]
ft^3	m^3			
<3	<0.085	5	Square or rectangular cube	Located at four corners and one located at the approximate center (see Figure 1).
			Cylindrical	Four located at the periphery 180 degrees apart on each end and 90 degrees apart from end to end and one located at the approximate center (see Figure 1).
≥3 and <225	≥0.085 and <6.4	9	Square or rectangular cube	Located at the eight corners of the cube and one located at the approximate center.
			Cylindrical	Three located at the periphery of each end, 120 degrees apart. One located at the approximate center, and the other two located to best represent the qualified work zone.
≥225	≥6.4	See Table 18	Square or rectangular cube	Located at the eight corners of the cube and one located at the approximate center. Others uniformly distributed to best represent the qualified work zone.
			Cylindrical	Three located at the periphery of each end, 120 degrees apart. One located at the approximate center, and the others located to best represent the qualified work zone.

(1) The location of TUS sensors for ≥3 ft^3 to <225 ft^3 may be used for volumes <3 ft^3.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

Table 18 - Number of TUS sensors required for batch furnaces, salt baths, controlled temperature liquid baths, fluidized bed furnaces, or continuous furnaces surveyed using the volumetric method

Furnace Class	Qualified Work Zone Volume Less Than	3 ft^3 0.085 m^3	225 ft^3 6.4 m^3	300 ft^3 8.5 m^3	400 ft^3 11 m^3	600 ft^3 17 m^3	800 ft^3 23 m^3	1000 ft^3 28 m^3	2000 ft^3 57 m^3	3000 ft^3 85 m^3	4000 ft^3 113 m^3
1 and 2	Number of sensors	5	9	14	16	19	21	23	30	35	40
3 to 6		5	9	12	13	14	15	16	20	23	25
1 and 2	ft^3 per sensor	<1	25	21	25	32	38	43	67	86	100
3 to 6		<1	25	25	31	43	53	63	100	130	160
1 and 2	m^3 per sensor	<0.03	0.7	0.6	0.7	0.9	1.0	1.2	1.9	2.4	2.8
3 to 6		<0.03	0.7	0.7	0.9	1.2	1.5	1.8	2.8	3.7	4.5

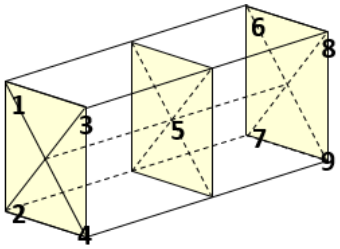
(1) For qualified work zone volumes greater than 4000 ft^3 or 113 m^3, the applicable furnace class formula as illustrated in the table below shall be used to determine the number of TUS sensors.

(2) For qualified work zone volumes less than 4000 ft^3 or 113 m^3, it is acceptable to use the appropriate furnace class formula, as illustrated in the table below or interpolation, to determine the number of TUS sensors.

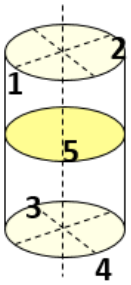
Furnace Class	Formula
1 and 2	9 + 1/2 X √(work zone volume in ft^3 - 225 ft^3) or 9 + 1/2 X √(35.3 X (work zone volume in m^3 - 6.4 m^3))
3 to 6	9 + 1/4 X √(work zone volume in ft^3 - 225 ft^3) or 9 + 1/4 X √(35.3 X (work zone volume in m^3 - 6.4 m^3))

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris

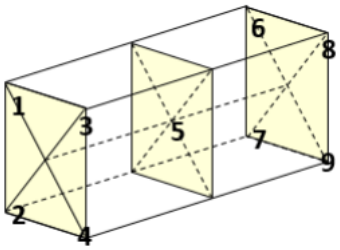
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

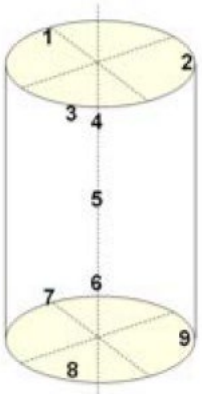


For square/rectangular volumes <3 ft^3 or 0.085 m^3, users shall choose locations 1, 4, 5, 7, and 8
or
locations 2, 3, 5, 6, and 9.
Location 5 being in the center.



For cylindrical volumes <3 ft^3 or 0.085 m^3, locations 1 and 2 and 3 and 4, the pairs being 180 degrees apart at the opposite ends, and location 5 in the center shall be used. Locations 1/2 and 3/4 shall be offset 90 degrees rotationally from each other.





For square/rectangular volumes and cylindrical volumes ≥3 ft^3 or 0.085 m^3 but less than 225 ft^3 or 6.4 m^3, all nine locations shall be used as shown. Cylindrical location 5 being in the center. Locations 4 and 6 are examples only.

Figure 1 - TUS sensor locations for square/rectangle cubic and cylindrical volumes

Once locations are established during initial TUS, the same locations shall be used during periodic TUSs (see 3.5.9.2).

Numbering of sensor locations in diagrams is provided as an example only.

3.5.9.2 For qualified work zone volumes ≥225 ft^3 or 6.4 m^3, the additional TUS sensors required by Table 18 shall be uniformly distributed to best represent the qualified work zone. One TUS sensor shall be located at the approximate center of the qualified work zone volume. When radiant heat from the periphery of the qualified work zone is used to heat the parts or raw material, the additional TUS sensors shall be uniformly distributed at the periphery of the qualified work zone. Once locations are established during the initial TUS, the same locations shall be used during periodic TUSs.

3.5.9.3 When a retort is used, the following shall apply:

3.5.9.3.1 The temperature of the furnace in which the retort is inserted shall be controlled so that the specified temperature is maintained within the retort. TUS sensors shall be located within the retort.

3.5.9.3.2 The tip (measuring junction) of at least one TUS sensor shall be as close as practical to the tip (measuring junction) of the sensor used to record temperature within the retort during production, but the sensor tip to tip distance shall not exceed 2 inches or 50 mm.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.10 TUS Data Collection

3.5.10.1 Data collection shall begin when the temperature of all TUS and furnace sensors are at least 100 °F or 55 °C below each TUS temperature so that any TUS or furnace sensor failing to reach the lower TUS tolerance or exceed the upper TUS tolerance can be detected. This is not applicable to salt bath or other thermal processing equipment that is maintained at a single set point. For TUS temperatures of 200 °F or 93 °C and below, data collection shall begin at the ambient temperature of the furnace or refrigeration equipment prior to the start of heating (or cooling for refrigeration equipment). If the furnace or refrigeration equipment is pre-stabilized, data collection shall begin prior to the TUS sensors being inserted.

3.5.10.2 Once data collection begins, temperature data shall be recorded simultaneously from all TUS sensors at a frequency of at least one set of readings every 2 minutes for the duration of the TUS.

3.5.10.3 Temperature data from control and recording sensors required by the applicable instrumentation type (see Table 9) shall also be recorded by the process recorder during the TUS. Load sensors used in addition to the identified instrumentation type during production are not required to be recorded during the TUS (e.g., if a Type D instrumentation furnace occasionally uses a load sensor, a load sensor is not required during the TUS), but those recording systems shall require instrument calibration and an SAT.

3.5.10.4 Regardless of the frequency of temperature data collection of control and recording sensors used during production, the temperature data shall be documented in the normal format used during production throughout the TUS.

3.5.10.5 If the normal frequency of control and recording sensor temperature data collection used during production is greater than 5 minutes, the recording frequency of these same sensors shall not exceed 5 minutes during the TUS.

3.5.10.6 At no time shall any TUS, control or recording sensor exceed the upper TUS tolerance. The equipment shall be held at the TUS temperature until all TUS sensors have stabilized (see 2.4.71). After stabilization, data collection shall continue for a minimum of 30 additional minutes.

3.5.10.6.1 For Unloaded/Empty Equipment

If the temperature readings of any TUS sensor exhibit an upward or downward trend that is not converging toward the set temperature, the data collection shall be extended as necessary until the trend is no longer evident.

3.5.10.6.2 For Equipment Tested with a Load

If a load is used during the TUS, some TUS sensors may continue to rise in temperature and slowly approach the set temperature. This rise in temperature of TUS sensors toward the set temperature meets the requirement for stabilization.

3.5.11 TUS Requirements for Continuous and Semi-Continuous Furnaces

3.5.11.1 Continuous and semi-continuous furnaces may be surveyed with TUS sensors arranged volumetrically or in a plane.

3.5.11.2 Both the volumetric and plane methods shall measure the entire qualified work zone volume. The difference between the two methods is the arrangement and number of TUS sensors.

3.5.11.3 Regardless of which method is used, the entire qualified work zone volume shall be surveyed. Multiple surveys may be required to accomplish measurement of the entire qualified work zone volume. For tubular furnaces equipped with several passages (e.g., ceramic tubes), each individual passage/tube used shall be surveyed.

3.5.11.4 All parameters used during the TUS shall reflect the normal operation of the equipment used during production.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.11.5 An initial TUS shall be performed at the minimum and maximum temperatures of the qualified operating temperature range(s) at the highest and lowest traverse speeds used during production. The additional temperatures of 3.5.2.2 apply to the initial TUS. The periodic TUS may be performed at any traverse speed used during production. A TUS shall be performed at the highest and lowest traverse speeds used during production at least annually.

3.5.11.6 Volumetric Method

3.5.11.6.1 TUS sensors shall be located in three dimensions to represent a portion, e.g., basket, tray, or the entire qualified work zone volume.

3.5.11.6.2 The number and location of TUS sensors shall be as stated in Figure 1 and Tables 17 and 18 based on the volume of the TUS basket or tray(s) used.

3.5.11.6.3 When surveying a portion of the qualified work zone incrementally, the entire volume of that portion shall be surveyed as the TUS sensors traverse through the furnace.

Table 19 - Number and location of TUS sensors for continuous and semi-continuous furnaces when using the plane method

Qualified Work Zone Height	Qualified Work Zone Cross Section	Minimum Number of TUS Sensors	Location of TUS Sensors
<1 foot or 300 mm	For circular section: radius <3 inches or <76 mm For rectangular section: width and height <6 inches or <152 mm	1	TUS sensor shall be at the center.
	Other furnaces with a qualified work zone <1 foot or 300 mm in height	Three with one additional sensor for each 2 feet or 610 mm of width over 8 feet or 2.4 m	Two TUS sensor locations shall be at the opposite sides of the qualified work zone at the approximate mid-height. One TUS sensor shall be at the approximate center. Additional TUS sensors shall be uniformly distributed throughout the plane perpendicular to the conveyance direction.
≥1 foot or 300 mm	<8 ft^2 or <0.75 m^2	5	Four TUS sensor locations shall be at the corners of the qualified work zone. One TUS sensor shall be at the approximate center. The remaining TUS sensors shall be symmetrically distributed about the center of the plane perpendicular to the conveyance direction.
	≥8 ft^2 or ≥0.75 m^2 and ≤16 ft^2 or ≤1.5 m^2	7	
	>16 ft^2 or 1.5 m^2	9	

3.5.11.7 Plane Method

3.5.11.7.1 TUS sensors shall be located in a single plane perpendicular to the furnace conveyance direction such that passing the plane through the furnace measures the entire qualified work zone volume of all zones required to be surveyed.

3.5.11.7.2 The number and location of TUS sensors shall be as stated in Table 19 based on the height and cross section of the qualified work zone.

3.5.11.7.3 A TUS sensor for each location shall be secured to a rack or in a load and traversed through the qualified work zone.

3.5.11.7.4 All required locations need not be traversed simultaneously; multiple TUSs may be required to accomplish measurement of the entire qualified work zone volume.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.11.8 TUS Data Collection

3.5.11.8.1 Temperature readings from all TUS sensors shall be recorded at least every 2 minutes with a minimum of ten sets of readings recorded for each qualified work zone. The traverse may be repeated as many times as necessary to ensure that any recurrent temperature pattern is identified at all locations through each qualified work zone. If the normal frequency of control and recording sensor temperature data collection used during production is greater than 5 minutes, the recording frequency of these same sensors shall not exceed 5 minutes during the TUS.

3.5.11.8.2 Starting the TUS with the furnace temperature higher than the TUS temperature is not permitted unless:

It is performed in only the initial or preheating zones of multi-zone furnaces.

OR

It is specifically permitted by the applicable material or process specification.

3.5.11.9 The qualified work zone length is a unique value for each traverse speed and shall be calculated and documented after the TUS. Each qualified work zone length is the sum of the elapsed time during which the TUS sensors were within the required TUS tolerance at the traverse speed used.

3.5.11.10 There shall be objective evidence that the soak time and temperature conform to the process or material specification requirements using the stated traverse speed. The qualified work zone length is the traversed distance through which all TUS sensor readings meet the required TUS tolerance. --`,,`,,,,`,,`,`,,`,`,,`,,`,,`,`,,`,`,,,,`-`-`,,`,,`,`,,,,---

3.5.12 Alternative TUS Methods for Continuous, Semi-Continuous Furnaces, or Furnaces with Retorts or Muffles

3.5.12.1 Where it is impractical to traverse the TUS sensors through a continuous or semi-continuous furnace, or to install TUS sensors into the retort, batch, or muffle of a furnace, it is acceptable to use either the probe method (see 3.5.12.2) or a property survey (see 3.5.12.3).

3.5.12.2 Probe Method

3.5.12.2.1 In lieu of locating TUS sensors as stated in Figures 1 and Table 19, it is acceptable to insert TUS sensors through the side walls, hearth, or roof within 3 inches or 76 mm of the locations identified in Figure 1 and Tables 17 and 18. If this method is used, the number of TUS sensors shall be as stated in Figure 1 and Tables 17 and 18 based on the volume of the qualified work zones surveyed.

3.5.12.2.2 Temperature readings from all TUS sensors shall be taken at least every 2 minutes for a minimum of 30 minutes once all sensor temperatures have stabilized. If the normal frequency of control and recording sensor temperature data collection used during production is greater than 5 minutes, the recording frequency of these same sensors shall not exceed 5 minutes during the TUS.

3.5.12.2.3 If a continuous or semi-continuous furnace is being probed, it is not required that a load be traversed through the furnace during the TUS.

3.5.12.3 Property Survey

3.5.12.3.1 A property survey shall require:

Testing material initially and annually thereafter.

AND

Monthly analysis of property survey trends. The material selected shall be one whose properties are known to be sensitive to variations in thermal processing soak times and temperatures and, whenever possible, one that is processed frequently.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.12.3.2 Material thickness shall be within the normal size range used during production. If a two-step treatment is required (e.g., harden and temper), it is permitted to perform the second step on the test specimens separately from the remainder of the lot, e.g., in a laboratory furnace.

3.5.12.3.3 Initial and annual property surveys shall be performed at the highest and lowest operating temperatures used during production. Additional temperatures shall be added to ensure that no two adjacent temperatures are greater than 300 °F or 165 °C apart. Continuous and semi-continuous furnace traverse speeds shall be those normally used during production. At least ten test specimens shall be processed at each operating temperature.

3.5.12.3.4 Test specimens shall be taken from parts or raw material located at the extremes and center of the load except for coils, where test specimens shall be taken at both ends of the coil. Coil test samples shall be tested at both edges and at the center of each specimen.

3.5.12.4 Monthly Analysis of Property Survey Trends

3.5.12.4.1 Properties of thermally processed specimens shall be analyzed by a statistical technique described in ASTM MNL7 or other recognized statistical process control reference work.

3.5.12.4.2 If the trend of properties exhibits a shift outside of the known upper or lower control limits, no further processing shall occur until the cause of the shift is determined and corrected. The requirements of Section 4 shall apply.

3.5.13 TUS Sensor Failure Requirements

3.5.13.1 TUS sensor or recording channel failure at the corner locations of a rectangle/square work zone or periphery (top/bottom or front/back) locations of a cylindrical work zone is not permitted. A temporary condition such as a short or loose connection where normal temperature indication is restored shall not be considered as a failed TUS sensor.

3.5.13.2 The failure of one or more TUS sensors at corner locations, any two adjacent TUS sensors, or a number exceeding the requirements in Table 20 or Table 21 shall require corrective action to be taken and the TUS repeated.

Table 20 - Allowable number of TUS sensor failures for TUSs performed <2000 °F or <1093 °C

Total Number of TUS Sensors	Allowable Number of TUS Sensor Failures
3 to 9	None
10 to 16	1
17 to 23	2
24 to 39	3
≥40	No more than 10%

Table 21 - Allowable number of TUS sensor failures for TUSs performed ≥2000 °F or ≥1093 °C

Total Number of TUS Sensors	Allowable Number of TUS Sensor Failures
3 to 5	None
6 to 9	1
10 to 16	2
17 to 23	3
24 to 39	4
≥40	No more than 10%

--`,`,,,,``,,`,`````,,``,,`-``,,`,,,,-```,,`,,,,

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris

Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.14 TUS Pass/Fail Requirements

3.5.14.1 A TUS shall be considered acceptable if all previously described requirements are met, including the following:

3.5.14.1.1 When compiling and analyzing TUS temperature data, compensation for known correction factors for the TUS sensors and TUS instrumentation shall have been applied algebraically.

3.5.14.1.2 Control and recording sensor readings and corrected TUS sensor readings shall not have exceeded the upper TUS tolerance at any time.

3.5.14.1.3 The time required to achieve TUS sensor temperature recovery, stabilization, or a recurrent temperature pattern shall not have exceeded the time limit specified in the applicable material or process specification.

3.5.14.1.4 The TUS soak period shall not have been less than the minimum required time.

3.5.14.1.5 The readings of all TUS sensors and control or recording sensors shall have been within the temperature tolerance requirements shown in Table 15 or Table 16 during the minimum TUS soak period except as allowed by 3.5.13.

3.5.14.1.6 If any temperature overshoot occurred where the upper TUS tolerance was exceeded during the approach period or the minimum TUS soak period, the temperature overshoot shall have been noted on the TUS report and the TUS shall have been documented as a failed TUS.

3.5.14.1.7 If the TUS was not within the allowable tolerances of Table 15 or Table 16, the cause of the deviation shall be determined and documented. The requirements of Section 4 apply.

3.5.14.1.8 For equipment surveyed at an extended periodic TUS interval, a TUS failure shall be cause for the interval to revert to the normal periodic TUS interval specified in Table 15 or Table 16 until the required number of consecutive successful periodic TUSs are completed.

3.5.14.1.9 If the corrective action for a failed TUS takes the form of implementing a modification offset, and if the qualified operating temperature range exceeds 300 °F or 165 °C, a resurvey is required where uniformity is verified at the temperature extremes of the TUS range where the offset is applied. TUS temperatures for each range where offsets are applied shall not be more than 600 °F or 335 °C apart.

3.5.15 Relocation of Hot or Cold Recording Sensors for Type A and C Instrumentation

3.5.15.1 When the hot and cold temperature locations change within the furnace based on the readings from the most recent TUS, the hot and cold recording sensors may need to be relocated to reflect the new hot and cold locations within each qualified work zone.

3.5.15.2 The hot and/or cold recording sensors do not require relocation if either of the following conditions is met:

3.5.15.2.1 The temperature uniformity results do not exceed one half of the maximum temperature uniformity tolerance for the applicable furnace class at all temperatures surveyed. The intent of this condition is that the TUS results do not exceed 1/2 of the uniformity tolerance in the cold direction from the set point temperature to evaluate the cold recording location and that the TUS results do not exceed 1/2 of the uniformity tolerance in the hot direction from the set point temperature to evaluate the hot recording location. The hot and cold locations are to be assessed individually.

Example: Furnace Class 2 ±10 °F and the TUS results showed that both the hot and cold locations changed. The TUS results were -4.0 °F to +5.5 °F. This means that the cold location does not need to be relocated, but because the TUS results exceeded the half tolerance in the positive direction, the hot location does not pass this test.

NOTE: A TUS modification offset may be used to center the TUS results to meet the above requirement for both locations (see Figure 2).

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

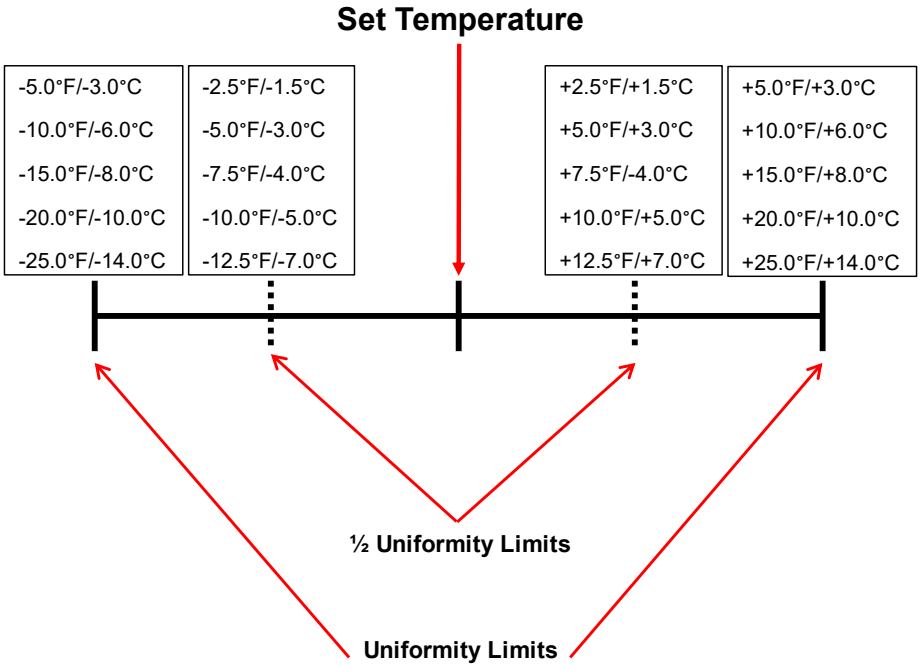


Figure 2 - Hot/cold sensor relocation condition #1

3.5.15.2.2 The difference between the measured temperature at the current hot and/or cold recording sensor locations and the actual respective hottest and coldest measured locations does not exceed the maximum SAT difference for the applicable furnace class (see Tables 11 and 12). The intent of this condition is that the maximum and minimum TUS sensor readings during the 30-minute TUS soak, recorded and corrected by the TUS recorder, are to be compared to the furnace recording of the current hot and cold recording sensors as close to the same time as possible given the data collection rates of the furnace and field test recorders and shall not exceed the applicable SAT difference. See Figure 3.

Example: The cold position passed the first test, but the hot position did not pass the first test. Therefore, the hottest corrected TUS temperature is compared to the temperature of the current furnace hot location at the time. If the difference is ≤ the Max SAT difference, then the hot location does not need to be relocated.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

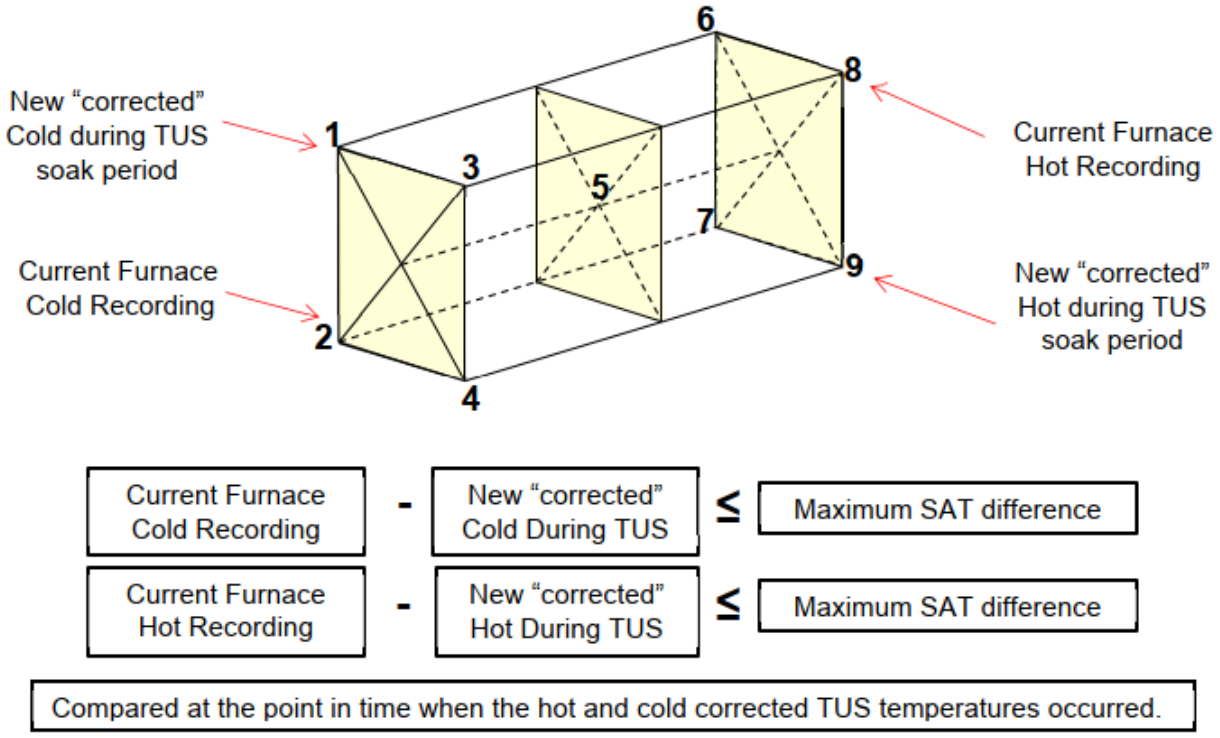


Figure 3 - Hot/cold sensor relocation condition #2

3.5.15.3 If the qualified operating temperature range exceeds 300 °F or 165 °C and the hot and/or cold recording sensors are permanently positioned/fixed and need to be relocated, additional TUS temperatures shall be performed to verify that the new locations are correct throughout the qualified operating temperature range of the furnace.

3.5.15.4 If the hot and/or cold recording sensors are not permanently positioned/fixed, they can be relocated without the above testing if there is a diagram of the required location where the hot and cold recording sensors will be placed at each process temperature.

3.5.16 TUS Results and Records

3.5.16.1 The results of the TUS shall be documented. At a minimum, the TUS documentation shall include:

a. Furnace identification name or number.

b. Identification of TUS as initial or periodic.

c. TUS instrument unique identification number.

d. TUS sensor(s) batch or lot number.

e. TUS set point temperatures.

f. Control instrument tuning constants.

g. TUS sensor calibration report.

h. TUS instrument calibration report.

i. TUS, control and recording sensor location identification required by the applicable instrumentation type, including a detailed diagram, description, or photograph(s) of any load, rack, or fixture used.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

j. The atmosphere used.

k. Time and temperature data from all recorded sensors required for the instrument type for all qualified work zones surveyed.

l. Correction factors for the TUS sensors and TUS instrument at each TUS temperature. The TUS instrument correction factors shall be stated even when the correction factors have been electronically applied to the TUS instrument to correct the TUS instrument temperature readings.

m. As-found and as-left correction and/or modification offsets if used during production.

n. Corrected or uncorrected (if documented) readings of all TUS sensors at each TUS temperature. TUS readings shall be identified as corrected or uncorrected.

o. Load condition, i.e., empty, with a rack or with a load.

p. TUS start date and time (when temperature data collection began).

q. TUS end date and time (when temperature data collection ended).

r. The results of hottest and coldest TUS and furnace recording sensor relocation analysis for Types A and C instrumentation, as applicable.

s. Traverse speed(s) or shaker frequency and qualified work zone length for the TUS performed on continuous and semi-continuous furnaces, as applicable.

t. TUS pass or fail statement.

u. When applicable, documentation of any control, recording, or TUS sensor failures.

v. Summary of the hottest and coldest corrected TUS readings at each test temperature during the minimum soak period compared to the TUS requirement. For example, a TUS performed at 920 °F ± 10.0 °F or 493 °C ± 6.0 °C. During the 30-minute soak period, the hottest corrected temperature is 923.4 °F or 495.2 °C, and the lowest corrected temperature is 918.8 °F or 492.7 °C. The summary would be -1.2 to +3.4 °F or -0.3 to +2.2 °C.

w. Identification of the technician performing the TUS.

x. Identification of the agency if TUS is not performed internally.

y. Approval of an authorized agent for the calibration agency if not performed internally.

z. User quality organization approval.

3.5.17 Radiation Survey

When required by the applicable material or process specification, a radiation survey shall be performed as follows:

3.5.17.1 Where the heat source (e.g., electrical elements or gas radiant tubes) is located in the furnace walls, ceiling, or floor, and the heat source has either a direct line of sight to the work zone or is only separated from the work zone by a single sheet of metal, a radiation survey shall be performed at the maximum operating temperature used for aluminum during production. The radiation survey shall be performed initially and after any equipment repair or modifications, including furnace relocation, that could affect the radiation characteristics of the heat source.

3.5.17.2 The radiation survey sensor(s) shall be in addition to the required number of TUS sensors. The radiation survey and the initial or periodic TUS may be performed simultaneously or separately.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.5.17.3 The radiation survey sensors shall be peened into or welded onto the center of 6061 aluminum alloy panels. The panels shall measure approximately 12 x 12 inches or 30 x 30 cm and not more than 0.125 inch or 3.1 mm in nominal thickness. The panels shall have been heated in air to a temperature in the range of 970 to 1010 °F ± 10.0 °F or 520 to 545 °C ± 6.0 °C and air cooled prior to the first radiation survey. Soak time shall be in accordance with the thickness requirements for the solution treatment as defined by the material or process specification.

3.5.17.4 The panels, one for each 10 ft^2 or 0.93 m^2 of furnace side wall, ceiling, and/or floor area where the heat source is located, shall be distributed symmetrically, with the faces parallel to the furnace wall, ceiling, and/or floor where the heat source is located, at the outer limits of the qualified work zone. Either side of the panel(s) may face the heat source.

3.5.17.5 All radiation survey sensor readings shall meet the data collection requirements of 3.5.10.2, sensor failure requirements of 3.5.13, and the TUS requirements of 3.5.14.

3.5.18 TUS Interval Deviations

3.5.18.1 The user is allowed to have thermal processing equipment exceed the allowable TUS interval, including any extension days in Table 22, with the approval from each applicable cognizant engineering or quality organization. When the TUS is next performed, it shall include all required temperatures of the initial TUS and at the highest and lowest traverse speed for continuous and semi-continuous furnaces. The TUS interval shall continue at the interval in use at the time of the last periodic TUS.

3.5.18.2 For thermal processing equipment that has been documented as being “out of use/service” beyond the next TUS due date and any extension days in Table 22, a periodic TUS shall be performed before being returned to service. The TUS interval shall continue at the interval in use at the time the equipment was taken “out of use/service”.

3.6 Laboratory Furnaces

3.6.1 General Laboratory Furnace Requirements

3.6.1.1 Laboratory furnaces shall be used for preparation of laboratory testing samples such as, but not limited to, response to heat-treatment testing in accordance with the material or process specification.

3.6.1.2 A laboratory furnace shall not be used for thermal processing of any parts or raw material unless the furnace meets all applicable requirements of this specification.

3.6.2 Laboratory furnace requirements when a load sensor is used include:

a. Load sensors shall comply with 3.1.10.

b. Calibration of control and recording instruments shall be performed quarterly. Calibration shall meet the requirements of Table 7.

c. The SAT shall be performed quarterly. The SAT difference shall meet the requirements of Table 11 or Table 12 for the lowest furnace class required by the material or process specification specified.

d. After completion of a successful initial TUS plus two consecutive successful periodic quarterly TUS, the TUS interval may be extended to semiannually. The TUS results shall meet the requirements of Table 15 or Table 16.

3.6.3 Laboratory furnace requirements when a load sensor is not used include:

a. Control and recording instrument calibration, SAT, and TUS shall be performed as required by the instrument type and furnace class applicable for production equipment.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

3.7 Records

3.7.1 All records of sensor calibration, instrument calibration, SAT, and TUS in addition to any calibration, SAT, and TUS failures shall be available for inspection and shall be maintained for not less than 5 years.

3.8 Rounding

3.8.1 If used, rounding shall be applied in accordance with a documented procedure and used in a consistent manner.

3.8.2 Rounding to 0.1 °F or 0.1 °C in accordance with ASTM E29, other equivalent international standards or commercial software programs is acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 The user shall be responsible for the performance of all required calibration and tests and for conformance to all applicable requirements specified herein. The purchaser reserves the right to witness calibration or tests specified herein to ensure that processing conforms to applicable requirements, but such witnessing shall not hinder any process within the facility.

4.2 Third-party (external) pyrometry service provider companies shall have a quality system accredited to ISO/IEC 17025 from an ILAC (International Laboratory Accreditation Cooperation) recognized regional cooperation body. The scope of accreditation shall include the laboratory standards and/or field service (for testing and calibration activities), as applicable. This applies to third-party providers performing calibrations or tests at the user's site or in their own laboratory.

4.3 When calibrations and/or tests defined herein are performed by the user's personnel, the user shall have procedures for calibration and testing, as applicable, detailing the methods and practices for the determination of the accuracy and measurements to minimize random errors. The user shall ensure the competence of personnel who perform calibrations and/or tests, evaluate results, and approve test results and documentation. Personnel who are undergoing training shall be appropriately supervised. Personnel performing specific tasks shall be qualified based on documented education, training, experience, and/or demonstrated skills, as applicable.

4.4 Equipment that is affected by any calibration or test failing to meet applicable requirements, or that has exceeded the allowable interval including any extension period defined in Table 22, shall be removed from service (i.e., production shall not begin once the extension days expire).

Table 22 - Permitted calibration/test interval extension

Interval	Allowable Extension Beyond the Date Due Calendar Days
Weekly	1
Biweekly	2
Monthly	3
Bimonthly	3
Quarterly	4
4 Months	4
Semiannually	6
Annually	12
Any interval greater than annually	12

4.5 As a result of any calibration or test failure, an evaluation of the possible effects of the failure on parts and raw material processed since the last successful corresponding calibration or test shall be performed.

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT

4.6 Appropriate corrective action shall be in accordance with established material review procedures, including the action taken to return the calibration or test to the required level of compliance and actions taken to prevent recurrence of the failure. The corrective action shall be documented and maintained on file.

4.7 A conforming corresponding calibration or test shall be required and documented before returning the equipment into service.

4.8 When parts or raw material processing conditions deviate from material or process specification requirements, the affected purchaser(s) shall be notified.

5. PREPARATION FOR DELIVERY

Not applicable.

6. ACKNOWLEDGEMENT

Not applicable.

7. REJECTIONS

Not applicable.

8. NOTES

8.1 Revision Indicator

A change bar (I) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document itle indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

8.2 Both imperial units and SI units are primary. Both Fahrenheit and Celsius temperatures are primary. Either the imperial/Fahrenheit system may be used or the SI/Celsius system may be used.

PREPARED BY SAE AMS B FINISHES PROCESSES AND FLUIDS COMMITTEE

Copyright SAE International
Provided by Accuris under license with SAE
No reproduction or networking permitted without license from Accuris
Licensee=Marshall of Cambridge Aero - Cambridge/9992343108, User=Claridge, David
Not for Resale, 03/17/2025 02:30:54 MDT